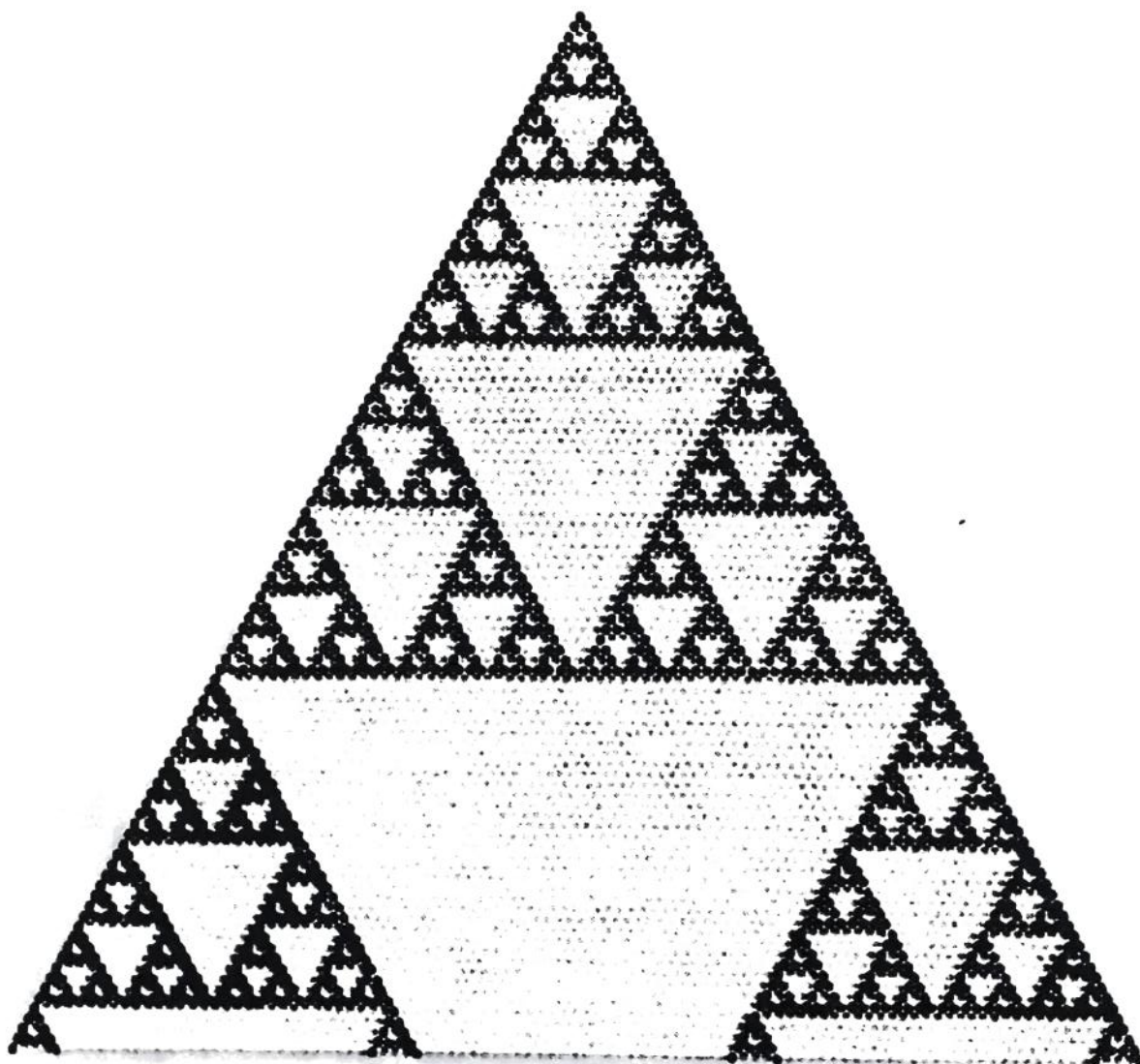


MATH SURVEY



STUYVESANT H.S. 1982

WE DEDICATE THIS ISSUE TO
THE STUYVESANT HIGH SCHOOL MATH TEAMS AND THEIR COACHES
PAST, PRESENT AND FUTURE.

The pattern depicted on the cover is a representation of the numbers in Pascal's Triangle by dots; the odd numbers are the black dots. For more on Pascal's Triangle, see p. 34 of this issue.

MATH SURVEY

Spring 1982

Vol. LVII

THE STUYVESANT HIGH SCHOOL

MATHEMATICS PUBLICATION

Edward Marcantonio
Assistant Principal
Department of Mathematics

Gaspar R. Fabbriante
Principal

TABLE OF CONTENTS

<u>Editorial Staff</u>	The Mailbox Principle.....Dana Randall	2
<u>Editors-in-chief</u>	Stewart's Theorem.....Ted Poon	4
Joel Hirschhorn	Math Team.....	5
David Zagorski	Coloring the 17-holed Torus....Joel Hirschhorn	6
<u>Associate Editors</u>	17.....	7
Larry Buxbaum	Games People Play.....Jean Lee	8
Nir Hacoheh	Perfect Numbers.....Jaehoon Chung	11
<u>Business Manager</u>	What is i^i ?.....Kevin Buehler	12
Jean Lee	Mass Point Math.....Keith Dienes	14
<u>Mathematical</u>	Fermat's Little Theorem.....David Yuen	16
<u>Consultant</u>	Fermat's Last Theorem.....David Zagorski	17
Noam Elkies	Infinite Descent.....Larry Buxbaum	18
<u>Layout Editors</u>	Regents Refresher.....	20
Kevin Buehler	Puzzle.....Jay Sipelstein	22
Keith Dienes	Math Society.....	23
Jeff Rosenfeld	What's a Fractal?!.....Jeff Rosenfeld	24
David Zuckerman	Westinghouse Tips.....	25
<u>Advertising Manager</u>	Wilson's Theorem.....Noam Elkies	26
Jeff Saltiel	Mathematical Miscellany.....	27
	Vector Proofs.....David Zuckerman	28
	Special Pythagorean Triples.....Kenneth Baron	30
	Problems.....Iuval Clejan	32
	Ptolemy's Theorem.....Enn-Ling Chen	33
	Pascal's Triangle...Mark Basch and Nir Hacoheh	34
	Answers.....	35
	Taxicab Geometry.....Anna Franco	38

Staff

Kenneth Baron	Ayame Konishi
Mark Basch	Wei Lam
Enn-Ling Chen	Vivian Pan
Jaehoon Chung	Ted Poon
Iuval Clejan	Dana Randall
Anna Franco	Jay Sipelstein
	David Yuen
<u>Faculty Advisor</u>	
Kurt E. Rittermann	

Thanks to:

The Hirschhorn family, for their hospitality and their support of Math Survey staff members, Mr. Glotzer, for the use of his typewriter, and especially Mr. Marcantonio, for his invaluable help.

The Mailbox Principle

by Dana Randall

Tommy went into math class one day, planning to impress his teacher with his new discovery. He went up to him and remarked, "Did you know that two people in New York City have the same amount of hair on their heads, excluding, of course, bald men?"

His teacher, confused by this outrageous statement, asked him inquisitively what he meant by it.

"Well," explained Tommy, "there are enough people in New York to set up a one-to-one correspondence between every person and every conceivable amount of hair a person could have on his head, and still have some people left over. Since each of these people must have one of the previously mentioned amounts of hair, there must be two people with the same amount of hair on their heads."

His teacher smiled, and was obviously impressed by this logic. "Tommy, you just used one-to-one correspondence, one of the most primitive forms of mathematics, in a very interesting manner. The idea you are suggesting is Dirichlet's drawer principle, often referred to as the pigeonhole or mailbox principle. This principle states that if you have ' m ' letters to put in ' n ' mailboxes, and $m > n$, then there will be at least one box with at least two letters in it. This is more or less the statement you just made. It is amazing how many difficult questions in mathematics can be reduced to the mailbox principle, which is so obvious and so simple."

"Give me an example," asked Tommy, who was now extremely fascinated by the subject and anxious to know more about it.

"Hm, let's see. Ah yes. Choose any ten numbers between 1 and 100."

Tommy considered for a moment. "How about 17, 49, 13, 28, 9, 85, 55, 34, 22, and, uh..., 97."

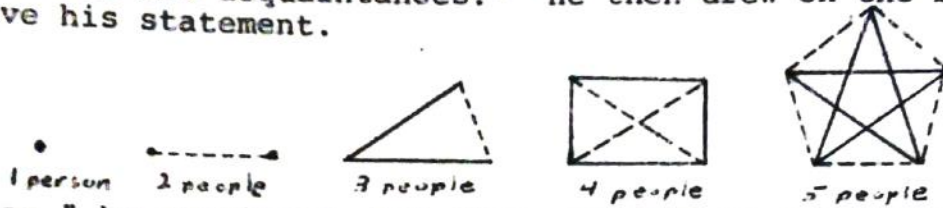
"Fine. I can choose two groups of these numbers that have the same sum. For example, $17 + 13 + 55 = 85$, or $28 + 85 = 49 + 9 + 55$. In fact, I can always choose two such groups no matter which ten numbers are picked. Do you know why?" Tommy shook his head. "Well, there are $2^{10} = 1024$ groups of numbers, and the sums of these groups are positive integers ranging up to $10 \cdot 100 = 1000$. Thus, there are more groups than possible sums, so two of the groups must have the same sum."

Tommy looked up. "That's much simpler than I expected! Can you think of any more?"

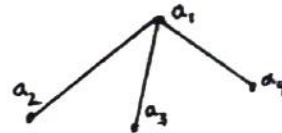
"Sure. I remember this question from when I was in school. 'What is the maximum number of integers one can choose so that neither the sum nor the difference of any two is divisible by 100?' The answer is 51, and this is how we can figure it out. We can write the possible remainders when any number is divided by 100 as $(-49, -48, \dots, -1, 0, 1, \dots, 49, 50)$. Two numbers with the same absolute value, modulo 100, have the property that their sum (if they have opposite signs) or difference (if they have the same sign) is divisible by 100. The absolute values can have only 51 different values $(0, 1, \dots, 49, 50)$; therefore the answer is 51."

Tommy was again impressed by the simplicity of his teacher's solution. "One more?"

"O.K.," replied his teacher, "but this is the last one. I have to start today's lesson soon. This is another disguised application of the mailbox principle. If there are five or fewer people, there do not have to be three mutual acquaintances or three total strangers amongst them. This can easily be shown with diagrams. Let each vertex be a person. A dotted line means they are strangers. A solid line means the two people they connect are acquaintances." He then drew on the blackboard to prove his statement.



"However," he continued, "if there are six or more people, there must be three mutual acquaintances or strangers. (In other words, either a solid or a dotted triangle must be formed, using a diagram as on this page). This is also easily proven. Each person in this case must have some relationship with at least five other people. At least three of them, according to the mailbox principle, must be of one nature, either acquaintances or strangers. For our purposes, let's assume they are acquaintances. Therefore, we have an arrangement as such.



None of the people (a_2 and a_3), (a_2 and a_4) or (a_3 and a_4) could be acquaintances also, or there will be three mutual acquaintances, (a_1 , a_2 , and a_3), (a_1 , a_2 , and a_4), or (a_1 , a_3 , and a_4), respectively. Therefore, a_2 , a_3 , and a_4 must all be strangers. In this case we have three mutual strangers.

"Now consider this problem when there are three possibilities. For example, they could be friends, strangers, or enemies. In this case, how many people must there be in order to make sure there will be three mutual friends, enemies, or strangers?"

Tommy thought for a minute. Finally he looked up. "Six... seventeen!" he exclaimed.

"Very good!" smiled his teacher. "I see you understand it well. Now class, for today's lesson. Did you know that there are two people in New York City with the same amount of hair on their heads?"



Stewart's Theorem

by Ted Poon

In 1745, Matthew Stewart published a geometric theorem which was later named after him. It described the relationship between an internal line segment of a triangle (a segment whose end-points are a vertex and a point on the opposite side, such as medians and altitudes), the two portions into which the segment divides the opposite side, and the three sides of the triangle.

Stewart's Theorem:

In triangle ABC , if $\overline{AB} = c$, $\overline{AC} = b$, $\overline{BC} = a$, and D is on $\overline{BC}$ such that $\overline{BD} = m$, $\overline{CD} = n$ and $\overline{AD} = d$ (see figure 1), then

$$a(mn + d^2) = mb^2 + nc^2.$$

Proof:

Applying the Law of Cosines to triangle ACD ,

$$\begin{aligned} b^2 &= d^2 + n^2 - 2dn\cos\theta; \\ b^2 - d^2 - n^2 &= -2dn\cos\theta; \\ d^2 + n^2 - b^2 &= 2dn\cos\theta. \end{aligned} \tag{1}$$

In triangle ABD ,

$$c^2 = d^2 + m^2 - 2dm\cos(180^\circ - \theta) \quad (\text{by the Law of Cosines}).$$

Since $\cos(180^\circ - \theta) = -\cos\theta$,

$$\begin{aligned} c^2 &= d^2 + m^2 - 2dm(-\cos\theta), \text{ or} \\ c^2 - d^2 - m^2 &= 2dm\cos\theta. \end{aligned} \tag{2}$$

Dividing (1) by (2), we get

$$\frac{d^2 + n^2 - b^2}{c^2 - d^2 - m^2} = \frac{2dn\cos\theta}{2dm\cos\theta};$$

$$\frac{d^2 + n^2 - b^2}{c^2 - d^2 - m^2} = \frac{n}{m};$$

$$md^2 + mn^2 - mb^2 = nc^2 - nd^2 - nm^2;$$

$$md^2 + nd^2 + mn^2 + nm^2 = mb^2 + nc^2;$$

$$(m+n)d^2 + mn(m+n) = mb^2 + nc^2.$$

But $a = m+n$, so

$$ad^2 + mna = mb^2 + nc^2, \text{ or}$$

$$a(mn + d^2) = mb^2 + nc^2.$$

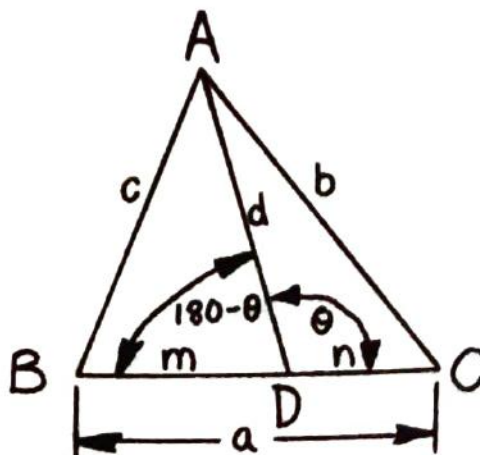


Figure 1

Q.E.D.

Math Team

The Stuyvesant High School Math Team, which meets every morning in rooms 515 and 517, has had another highly successful year in 1981-1982. Led by Coach Dr. Richard Rothenberg and co-captains Noam Elkies and Joel Hirschhorn, the team won every competition in which they entered. The senior team, which consists of Keith Dienes, Elkies, Hirschhorn, David Zagorski, and David Zuckerman, as well as alternates Robert Coyne, Paul Romanelli and Carl Weiner, won an easy victory over the other schools in New York City in the Fall 1981 Interscholastic Math League contests, and seems certain to win for the 17th consecutive term this spring. Junior team Red, led by captain Larry Buxbaum and perfect scorer Kevin Buehler, won in both Fall 1981 and Spring 1982 by the tremendous margin of one point. The other members were: Enn-Ling Chen, Iuval Clejan, Ted Poon, Jay Sipelstein and Darrell Tornay, and alternates Vance Berger and Naiyer Imam. Junior team Blue, which represented the efforts of captain Anna Franco, Kenneth Baron, Sui Ming Louie, Dana Randall and David Yuen, as well as alternates Mark Basch and Jaehoon Chung, performed admirably.

In the New York Math League competition, a statewide contest, perfect scorer David Zagorski has led Stuyvesant to victory in the first five of six meets. Barring a massive train or brain delay on the morning of the last meet, Stuyvesant should achieve victory in this area also. Everyone is invited to attend these meets, which are held at 8:00 AM in the West Cafeteria.

Stuyvesant did exceptionally well in the Annual High School Mathematics Exam (the MAA) with a team score of 415 (Noam Elkies, 145; Joel Hirschhorn, 140; David Zuckerman, 130), the highest in the city by 83 points and possibly the highest in the country. David Zagorski also had one of the top scores in the city with a 113 (making the national honor roll). The four students listed above were invited to participate in the 1982 U.S.A. Mathematics Olympiad (USAMO) which will be held on Tuesday, May 4th. In the 1981 USAMO, Noam Elkies and former captain Gregg Patrino tied for first, and represented 25% of the American team in the International Math Olympiad, where Elkies with a perfect score and Patrino with a second-place score helped to lead the United States team to victory.

The Math Team members performed outstandingly well in other areas. In the Westinghouse Science Talent Search, all five seniors on the senior team were semi-finalists, and the two co-captains were winners. Noam Elkies finished eighth in the country.

If you want to be a part of this winning tradition, you can come to rooms 515 and 517 at 8:00, where the members solve problems, take meets, learn methods for finding solutions to problems, and generally enjoy themselves. Everyone is welcome, especially beginners and freshmen. Join us!!

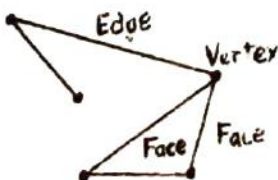
Coloring the 17-holed Torus

by Joel Hirschhorn

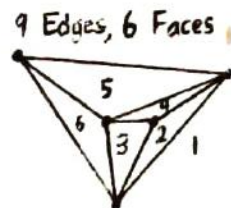
One of the most famous (and until recently, unsolved) problems in mathematics is the determination of the validity of the Four Color Map Theorem. The theorem states that given a map on a plane, it is possible to color the regions of the map in four colors or fewer so that no two regions sharing a common boundary are colored with the same color. This statement is equivalent to the following: Given a connected graph, it is possible to color the vertices of the graph in four colors or fewer so that no two vertices connected by an edge are colored with the same color. (A connected graph is a set of points, or vertices, connected by line segments, or edges, in such a way that none of the edges intersect and it is possible to trace a path from any vertex to any other vertex. The regions which are bounded by the edges are called faces.)



not a connected graph



connected graph

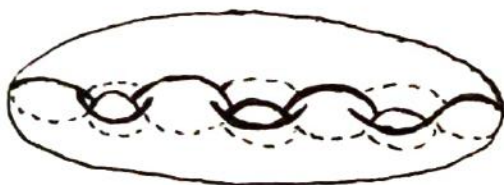


triangular connected graph

We will prove that in a k -holed torus, i.e. a donut with k holes, $k \geq 1$, it is possible to color the vertices of a connected triangular graph (that is, one in which twice the number of edges equals three times the number of faces) in the manner described above in N colors, where N is the positive root of the equation

$$N - 1 = 6 + \frac{12(k-1)}{N}. \quad (1)$$

To prove this theorem, we must derive the value of $\#Vertices - \#Edges + \#Faces$ for a k -holed torus. Euler proved that for a planar connected graph, $V - E + F = 2$. Below, the simplest connected graph on a k -holed torus is illustrated. We can see that the number of vertices is $2k+2$, the number of edges is $4k+2$, and the number of faces is 2. Thus, $V - E + F = 2k+2 - (4k+2) + 2 = 2-2k$.



Cut around the inside of each hole and then around the pieces joining the holes.

We will now prove the following lemma:

Lemma: The average number of edges emanating from each vertex is less than or equal to $N-1$, if $V > N$.

Proof: The average number of edges per vertex is $2E/V$ (since each edge emanates from two vertices). Since the graph is triangular, $2E=3F$, so

$$\frac{2E}{V} = \frac{3F}{V} = \frac{3(E-F)}{V}.$$

We know that $V-E+F=2-2k$, so $E-F=V+2k-2$. Substituting yields

$$\frac{2E}{V} = \frac{6(V+2k-2)}{V} = 6 + \frac{12(k-1)}{V}, \text{ which is less than or equal to}$$

$6 + \frac{12(k-1)}{N}$, since $V \geq N$ and $k-1 \geq 0$. But from (1), this last expression

equals $N-1$, so our lemma is proven.

Proof of theorem: We will proceed by induction on the number of vertices. If $V \leq N$, then we can color every vertex a different color. We now assume that every map with $j-1$ vertices is colorable with N colors, where $j \geq N$. By our lemma, in every j -vertex map there exists some vertex Q with $N-1$ edges or fewer emanating from it. If we remove Q and all of the edges leading from it, we have a map with $j-1$ vertices, which can be colored by the induction hypothesis. Suppose that Q is connected to vertices $V_1, V_2, \dots, V_{N-1}$. If these are colored with colors $C_1, C_2, \dots, C_{N-1}$, then we can color Q with color C_N . Thus the map is colorable with N colors. This theorem was proven by Haywood in the 1870's.

To calculate N for specific values of k , we can solve (1). Using the quadratic formula yields

$$N = \left\lfloor \frac{7 + \sqrt{1 + 48k}}{2} \right\rfloor = \text{the greatest integer less than or equal to the bracketed quantity.}$$

Substituting $k=17$ into this formula gives

$$N = \left\lfloor \frac{7 + \sqrt{1 + 816}}{2} \right\rfloor = \left\lfloor \frac{7 + 28.583\dots}{2} \right\rfloor = 17.$$

In other words, any triangular connected graph on a 17-holed torus can be colored in 17 colors or fewer!

17

The perceptive reader may have noticed 1 or 2 (or 17) references to the number 17 in this issue. In addition, it is possible that the reader has seen, on the 17th of every month, math team members wearing shirts with the number 17 (among other things) painted in yellow. This practice is due to exposure to a wonderful program, the Hampshire College Summer Studies in Mathematics Program, which takes place nearly every summer at, obviously, Hampshire College (in Massachusetts, not New Hampshire). This program is open to 60 or so mathematically gifted students, and is a great experience for learning, getting interested in math, and generally having a good time. What has this to do with 17? Well, 17 is the "sacred number" of the program. Why 17? For starters, it takes 17 colors to color a map on a 17-holed torus (see above), 17 is the number picked most often at random, 17 is the average of the first two perfect numbers, a regular 17-gon is constructible.... To learn many more properties while enjoying yourself immensely, write for an application to:

David C. Kelly (who "founded" 17)
Box SS
Hampshire College
Amherst, MA 01002

Games People Play

by Jean Lee

Little Janey scratched her head in puzzlement and squirmed in the high-backed chair. She was having a rather difficult time deciding which of Professor Riemann's soldiers she should try to capture without losing too many of her own soldiers.

Her opponent, Professor Georg Friedrich Bernhard "Bud" Riemann, patiently awaited her move and casually munched on some of the chocolate M & M's that they were using as stakes in their game. He had finally found the time (when he wasn't doing things like saving the universe or figuring out how far people with 4-inch foreheads could see*) to invite Little Jimmy and his younger sister, Little Janey, for a relaxing lunch and a strategy game in his study.

A sigh of impatience emanated from the nearby couch. "It always takes you forever to make one of these moves!" complained Little Jimmy, who was practically enveloped by the over-cushiony couch.

"It does not!" replied a defiant Little Janey.

"Calm yourself, Jimmy," said the professor. "Give the little lady a chance to think."

Little Jimmy groaned.

"Have some of these." Professor Riemann offered some of the chocolate M & M's to Little Jimmy.

As Little Jimmy strolled to the table to accept the M & M's, he examined his sister's position on the game board. He leaned over the table and mumbled something which apparently pertained to the game into Little Janey's ear.

"No coaching from the audience," requested Professor Riemann.

"But this game is going to go on forever if nobody helps her!" protested Little Jimmy.

"Very well," replied the professor in his usual erudite manner. "We'll give her a hand." He paused to munch pensively on an orange M & M. "She should capture my red soldier with her green one and surround the right side of my army."

"But how did you know so quickly?" chorused Little Jimmy and Little Janey.

"Why, Jim — that was quite an elementary problem! You should have been able to figure it out!" exclaimed a somewhat annoyed Professor Riemann. He was very much surprised and appalled that Little Jimmy could not solve the problem through his own efforts.

"But I don't understand how to solve it," persisted Little Janey meekly.

The professor had forgotten that Little Janey was not quite up to par with her brother's mathematical skills because Jimmy had had a head start of a few years. "All right," sighed Professor Riemann, "I'll explain the hows and whys of the solution for your little sister."

"And for Stuyvesant's 1982 Math Survey," Little Jimmy added.

*See the 1981 and 1980 editions of Math Survey, respectively.

Professor Riemann proceeded to draw the following diagram on a piece of paper:

		Bud		
		A	B	
Little Janey	A	2	-5	-5*
	B	7	-6	-6
Minmax(Bud)		7	-5*	

"What we have here," said the professor, "is a 2x2 matrix which displays each of the two strategies that both Janey and I have and the respective payoffs that each of us would — or would not — receive if each of us followed a certain strategy. You see, Janey has the option of either going after my red soldier with her green soldier and attacking my right side (we'll call that Janey's strategy A) or making a frontal assault with her blue tank by taking my yellow tank (strategy B). If she chooses strategy A, she'll have a longer but somewhat safer path to capturing my military base, which, of course, is the whole object of this game. Janey's strategy B presents a shorter path to my military base, but she'd have to have a little more luck to reach it."

"What are your two strategies?" inquired Little Jimmy.

"Well, I can retaliate by using either strategy A, which is to directly avenge any capture she makes of any of my pieces, or strategy B, which is to fortify my military base.

"The numbers in this game matrix show how much money — or M & M's in this case — one of us would receive if we followed a certain strategy. For example, if I followed my strategy A and Janey followed her strategy B, she would win 7 M & M's at the end of the game for following that strategy."

"I get it," said Little Jimmy. "The positive numbers correspond to the number of M & M's Janey would win, and the negative numbers represent the number of M & M's you'd receive after following a certain strategy."

"Exactly," continued Professor Riemann. "Now, in any game, your ultimate goal is to maximize your winnings, or minimize your losses. In order to do this, we've each written down the worst we can do by using each strategy. For example, by using strategy A, I can lose as many as 7 M & M's. Since each of us wish to minimize our losses, we take the more favorable of the two 'worst cases'. In Janey's case, this is the maximum of the minimums of the rows (the maxmin), or the maximum of -5 and -6, which is -5. In my case, we use the minimum of the maximums of the columns, or the minimum of 6 and -5, which is -5. (This is called the minmax.) Thus, Janey knows that by using strategy A, she cannot lose more than 5 M & M's, and I know that by playing my strategy B, I cannot win less than 5 M & M's. When the minmax equals the maxmin, as it does in this case, the game is said to have a 'saddle-point,' and both players should use only those strategies corresponding to the saddle-point. In other words, Janey should take my red soldier, and I should fortify my military base."

"But what if the game has no saddle-point?" queried Little Jimmy.

"In this case," explained the professor, "the players should employ a mixed strategy, which consists of mixing each of the pure strategies (such as strategy A or B) in certain proportions, or oddments. If a player's oddments are $p:q$, then he/she should play strategies A and B with frequencies in the ratio of $p:q$. We can use the minmax theorem to find the oddments of each strategy." He drew the following diagram:

		I	
		A	B
II	A	a	c
	B	b	d

Let this be any 2×2 game matrix:

oddments for I: $\frac{\text{Strategy A}}{\text{Strategy B}} = \frac{|c-d|}{|a-b|}$

oddments for II: $\frac{\text{Strategy A}}{\text{Strategy B}} = \frac{|b-d|}{|a-c|}$

"However, we're not finished yet," continued Professor Riemann. "In order to make the game absolutely fair, we must find the value of the game, which is the average win/loss for several plays of the game, assuming both players use their optimum strategies. If there is a saddle-point, the value of the game is equal to the saddle-point. Otherwise, the value of the game is defined as:

$$\frac{P(IA, IIA)P(IB, IIB) - P(IA, IIB)P(IIA, IIB)}{P(IA, IIA) - P(IA, IIB) - P(IIA, IIB) + P(IIA, IIB)} = \frac{ad-bc}{a-b-c+d}, \text{ where}$$

$a=P(IA, IIA)$ is the payoff when strategies IA and IIA are used, etc. If the value of the game is zero, the game is a fair one. However, if the value of the game is positive, the player listed to the left of the matrix will have winnings approximately equal to the value of the game. This player must pay the other player the value of the game before each round in order to keep the value of the game at zero. Of course, the opposite is true if the value of the game turns out to be negative."

Little Janey looked at the professor in wide-eyed amazement. "Gee, I didn't know that there was so much stuff on games!"

"Well," said the professor, "basic Game Theory — which I've just explained to you — isn't confined to games. It can also be applied to military strategies and to everyday-life situations."

"Enough Game Theory for the afternoon. Anybody for the mathematics of chocolate M & M's?"

WITH COMPLIMENTS FROM:



NEW YORK UNIVERSITY
COURANT INSTITUTE OF
MATHEMATICAL SCIENCES

Perfect Numbers

by Jaehoon Chung

Perfect numbers are essential elements in all numerological speculations. The world was created in 6 days, a perfect number. The moon orbits the earth in 28 days, again a symbol of perfection. We say that a number N is perfect if the sum of all its proper divisors, i.e. the divisors less than the number itself, equals N .

Perfect numbers were a favorite topic in number theory among the Ancient Greeks. One of their major results was stated by Euclid as follows: "If as many numbers as we please beginning from a unit are set out continuously in double proportion, until the sum of all becomes prime, and if the sum multiplied into the last make some number, the product will be perfect." This is stated in the following theorem:

Theorem: $2^{p-1}(2^p-1)$ is a perfect number if and only if 2^p-1 is prime.

(Prime numbers of the form 2^p-1 are called Mersenne primes. Some values of p which make this expression prime are 2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 127 and 257.)

Proof: We will make use of the formula used to calculate the sum of the divisors of N , $\sigma(N)$. It is known that N has a unique prime fac-

torization, which is of the form $p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$, where for $1 \leq i \leq r$, p_i and a_i denote a certain prime and the largest power of that prime dividing N , respectively. According to the formula,

$$\sigma(N) = (1+p_1+p_1^2+\dots+p_1^{a_1})(1+p_2+p_2^2+\dots+p_2^{a_2})\dots(1+p_r+p_r^2+\dots+p_r^{a_r}). \quad (1)$$

Recognizing that each parenthesized expression is the sum of a finite geometric series, we can rewrite (1) as follows:

$$\sigma(N) = \left(\frac{p_1^{a_1+1}-1}{p_1-1}\right)\left(\frac{p_2^{a_2+1}-1}{p_2-1}\right)\dots\left(\frac{p_r^{a_r+1}-1}{p_r-1}\right). \quad (2)$$

Since $\sigma(N)$ also includes N , which is not a proper divisor, we need to show that

$$\sigma(N) - N = 2^{p-1}(2^p-1).$$

From (2), when 2^p-1 is prime,

$$\sigma(N) = \left(\frac{2^p-1}{2-1}\right) \cdot \left(\frac{(2^p-1)^2-1}{(2^p-1)-1}\right) = (2^p-1)(2^p).$$

Thus,

$$\sigma(N) - N = (2^p-1)(2^p) - (2^{p-1})(2^p-1) = 2^{p-1}(2^p-1).$$

Now we need to show that this result only holds when (2^p-1) is prime. But this is trivial because when 2^p-1 is composite, $\sigma(2^p-1) > 2^p$, and since (2^{p-1}) is unchanged,

$$\sigma(N) > (2^p-1)(2^p), \text{ which implies that}$$

$$\sigma(N) - N > 2^{p-1}(2^p-1).$$

This proves our theorem.

What is i^i ?

by Kevin Buehler

In our study of mathematics, we soon encounter exponents. At first, we consider only positive integers; x^3 is defined as $(x)(x)(x)$. We then expand the definition to include all integers, so that $x^{-4} = 1/x^4$. Next, all rational numbers are considered,

and we define $x^{(a/b)}$ as $\sqrt[b]{x^a}$. Irrational powers can be considered as the limit of a series of approximating rational values. Even complex bases are included, such as $i^{17}=i$ or $(2+i)^2=3+4i$. Rarely, however, is an expression such as i^i defined. Evaluating i^i is the goal of this article.

First let us study some basics. Let us define e^x (where e is approximately 2.71828) as $1 + x + x^2/2! + x^3/3! + \dots + x^n/n! + \dots$ (this is the Maclaurin expansion).

Likewise, we let

$$\begin{aligned}\sin x &= x - x^3/3! + \dots + (-1)^n x^{2n+1}/(2n+1)! + \dots \\ \text{and } \cos x &= 1 - x^2/2! + \dots + (-1)^n x^{2n}/(2n)! + \dots\end{aligned}$$

where angles are expressed in radians. After careful scrutiny, we notice that

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

This allows us to represent any complex number $a+bi$ in the polar form $re^{i\theta}$, where r , the magnitude, equals $\sqrt{a^2+b^2}$ and $\tan \theta = b/a$.

Since a rotation of 2π is the same as none at all, $e^{i\theta}$ is periodic, and $re^{i\theta} = re^{i(\theta+2k\pi)}$, where k is any integer. To take the n th root of $a+bi$, we have

$$\sqrt[n]{a+bi} = (a+bi)^{1/n} = \left(re^{i(2k\pi+\theta)} \right)^{1/n}, \text{ or}$$

$$r^{1/n} e^{i(2k\pi+\theta)/n}, \text{ where } k \text{ is}$$

an integer, so we obtain n distinct n th roots (one for each value of k from 0 to $n-1$), each $\sqrt[n]{r}$ from the origin.

They form a regular n -gon inscribed in a circle of $\sqrt[n]{r}$ radius. This is actually a special case of De Moivre's Theorem, which states that if $u=r(\cos \theta + i \sin \theta)$ then $u^n = r^n (\cos n\theta + i \sin n\theta)$.

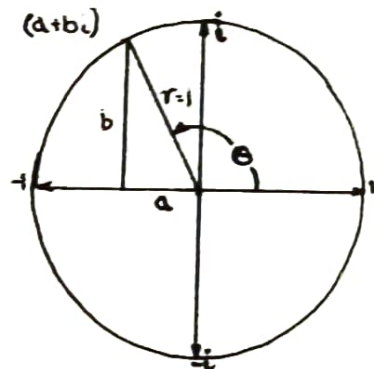
Let us get back to our original problem, evaluating i^i . To simplify our task, we will consider only those complex numbers that lie on the unit circle. These numbers are expressible as $e^{i\theta}$, since $r=1$. Let us raise $e^{i\theta}$ (which we set equal to w) to the $a+bi$ power.

$$w^{a+bi} = w^a (w^{bi}) = w^a e^{i(\theta+2k\pi)(bi)}. \quad (\text{By substitution for } w)$$

Simplifying, we get

$$w^{a+bi} = w^a e^{-b(\theta+2k\pi)} = \frac{w^a}{e^{(b\theta) e^{2k\pi b}}}, \text{ which we represent as}$$

the principal value of $\frac{w^a}{e^{b\theta}}$, and a geometric series of alternative



values, with a ratio of $e^{2\pi b}$. Note that geometrically, this represents points along a ray from the origin, as opposed to the points on a circle encountered when we used real numbers as exponents. We are now in a position to evaluate i^i .

$$w = i = e^{i\pi/2},$$

$$\theta = \pi/2,$$

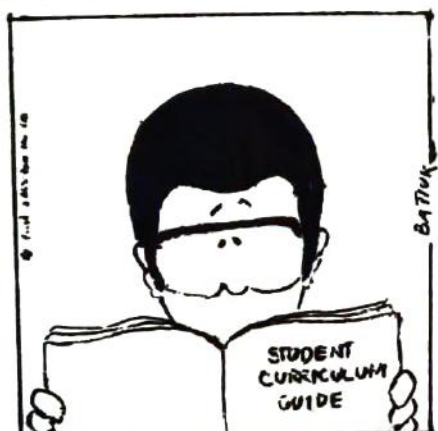
$$a = 0,$$

and $b = 1$, so

$$i^i = \frac{i^0}{e^{i\pi/2} \cdot e^{2k\pi}} = \frac{1}{e^{i\pi/2} \cdot e^{2k\pi}} \doteq .2079/(535.5)^k, \text{ where } k \text{ is an integer.}$$

Thus, we have discovered that i^i is always a real number. It might be interesting to expand the formula obtained above, modifying it so that any complex number, not just those lying on the unit circle, may be used as a base.

FUNKY WINKERBEAN



ALGEBRA II -

ALGEBRA II IS JUST LIKE ALGEBRA I EXCEPT MORE SO! AS IF FINDING OUT WHAT X IS WASN'T ENOUGH TO KEEP YOU BUSY, THEY ALSO THROW IN THE LETTER Y!

SOME PEOPLE ARE EXCITED BY THIS SORT OF CHALLENGE, BUT MOST OF US RECOGNIZE IT FOR THE SHAM IT TRULY IS!

2-9

HOW MANY GIRLS?

Those kids made a
terrible noise!
Near to fifty in
all with their toys,
The square of the square
Of the girls who were there
Came to double the
cube of the boys.

TWO BOYS AND A GIRL

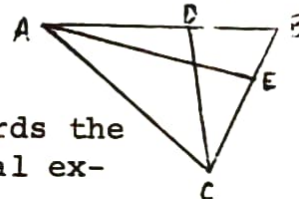
When Ted was twice
as old as Sue,
Her brother Clive
was twenty-two.
When Sue was twice
as old as Clive
Then Ted himself
was twenty-five.
Their ages total
one nought three,
So now what must
their ages be?

Mass Point Math

by Keith Dienes

"Mass point math" is a highly pragmatic method of problem-solving which has its foundations in physics and can be applied to geometry because of the strong inter-relationship that exists between math and physics. It acts as a favorable alternative to coordinate geometry methods which are often lengthy, difficult, and time-consuming.

Mass point math rests upon two basic properties: (1) that the sum of two points A (with assigned mass a) and B (with assigned mass b) is point C (with assigned mass $a+b$) on the line joining A and B such that $AC:CB = b:a$, and (2) that if, in a mass point diagram, two lines joining different sets of mass points intersect, they intersect in the point which is the sum of each of the original sets of points. These properties are derived from Archimedes' laws of balance and the "lever principle", and thus by assigning masses (or weights) to various points, we are in actuality balancing the triangle about a center. For example, if, in the figure below, $A=1$, $B=2$, and $C=1$, then D (the sum of A and B) has mass 3 and $AD:DB = 2:1$; then E (the sum of B and C) has the same mass and cuts BC similarly; and F is therefore the sum of both C and D or A and E.



To illustrate how one may apply "mass point math" towards the solution of a problem, let us consider the following general example:

PROBLEM: In triangle ABC, D is a point on $\overline{BC}$ and E is a point on $\overline{AC}$. $\overline{BE}$ and $\overline{AD}$ intersect in H. $\overline{CH}$ is extended to meet $\overline{AB}$ in F. If $\overline{DF}$ and $\overline{BH}$ intersect in G, find $BH:HE$ and $FG:GD$ if $BD:DC = 1:a$ and $CE:EA = 1:b$.

SOLUTION:

Step 1: Assign masses to the vertices of the triangle so that $D=B+C$ (i.e., the mass of D = the sum of the masses assigned to B and C) and $E=A+C$. For convenience, let $B=a$ and $C=1$. Since $CE:EA = 1:b$ and $C=1$, $A=(1/b)$. To avoid working with fractions, we multiply all the masses in the triangle by b , yielding

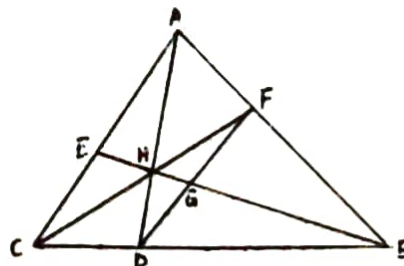
$$\begin{aligned} A &= 1 \\ B &= ab \\ C &= b. \end{aligned}$$

Step 2: Since $D=B+C$,

$$\begin{aligned} D &= ab + b. \text{ Similarly, } E = A+C \text{ implies that} \\ E &= b + 1. \text{ In addition, } H = B+E \text{ and } F = A+B, \text{ so} \\ F &= ab + 1 \end{aligned}$$

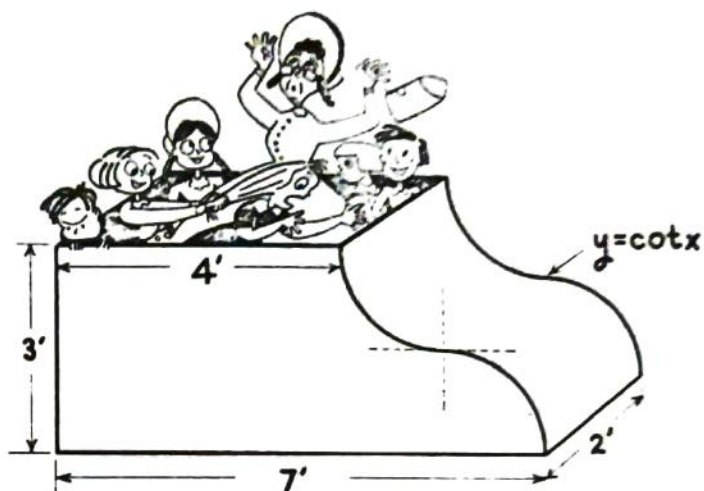
and $H = ab + b + 1$. Using these results, we determine that $G = 2ab + b + 1$.*

Step 3: At this point, the desired ratios can be calculated. From earlier results, $BH:HE = (b+1):ab$, $FG:GD = (ab+b):(ab+1)$, and the problem is solved.



Not every problem which may be solved using mass point methods is this general or involved. In fact, many problems which can be solved in this manner have one or two step solutions. Just remember, if you assign masses to all the points involved, and don't say the title of this article too many times in rapid succession, you'll be fine.

*Note that the masses of each of these points may be calculated in several different ways, each method yielding the same result. For example, $H = A + D$ as well, and since $A = 1$ and $D = ab + b$, $H = ab + b + 1$, as we have previously determined.



There was an old woman who lived in a shoe
Whose rectangular bottom scaled seven by two;
Rectangular top was two feet by four,
Height thirty-six inches and not a bit more;
Vamp a cotangent curve. — Find the volume within
And you'll know why the children were meager and thin.

YAACOV ROSENRAUCH
PRESIDENT

ALEPH LAMED YOD COMPUTERS

EDUCATION, CONSULTING AND SALES
2425 COYLE STREET
BROOKLYN N.Y. 11235

TELEPHONE
(212) 743-3294

LIC #

07638406155

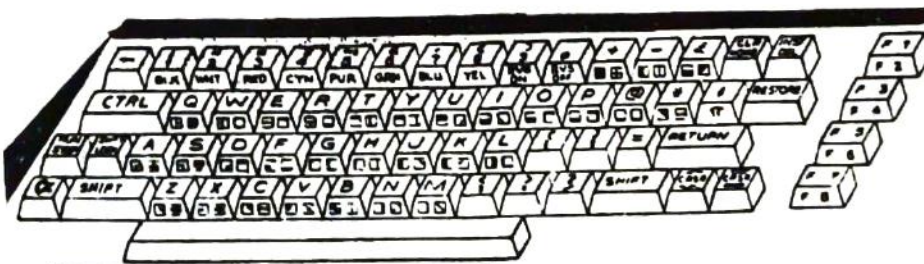
WOULD YOU BELIEVE
A COMPUTER FOR YOUR OWN HOME
AT AN UNBELIEVABLE PRICE

~~\$355~~ **REDUCED**
\$249
+tax
TRY BEATING
IT



VIC 28

tape recorder \$68



The VIC Keyboard

Fermat's Little Theorem

by David Yuen

Theorem: If p is a prime number, then the difference $a^p - a$ is, for any integer a , divisible by p .

Proof: Assume that a is not divisible by p . Then the integers $a, 2a, 3a, \dots, (p-1)a$ will also fail to be divisible by p , and they will yield different remainders when divided by p . If xa and ya ($p-1 \geq x > y$) produced identical remainders, then the difference $xa - ya = (x-y)a$ would be divisible by p . This is clearly impossible, since p is prime, a is not divisible by p , and $(x-y)$ is less than p . But since, upon division by p , all possible remainders are exhausted by the numbers $1, 2, 3, \dots, (p-1)$, it necessarily follows that

$$a = q_1 p + a_1,$$

$$2a = q_2 p + a_2,$$

$$3a = q_3 p + a_3,$$

$$\dots\dots\dots,$$

$$(p-1)a = q_{p-1} p + a_{p-1},$$

where $a_1, a_2, \dots, a_{p-1}$ are the positive integers $1, 2, \dots, p-1$, taken in some order. Multiplying together all these equalities, we obtain

$$1 \cdot 2 \cdot 3 \dots (p-1) a^{p-1} = Np + a_1 a_2 a_3 \dots a_{p-1};$$

$$1 \cdot 2 \cdot 3 \dots (p-1) a^{p-1} = Np + 1 \cdot 2 \cdot 3 \dots (p-1);$$

$$1 \cdot 2 \cdot 3 \dots (p-1) (a^{p-1} - 1) = Np.$$

Therefore it follows that $a^{p-1} - 1$ is divisible by p , and $a^p - a$ is also divisible by p . (If a is initially divisible by p , then the assertion of the theorem is obvious.)

Q.E.D.

Alternate Proof: The theorem is obvious for $a = 1$, since in that case, we have $a^p - a = 1 - 1 = 0$. We shall now apply mathematical induction to the positive integer a , and show that if $a^p - a$ is divisible by p , then it follows that $(a+1)^p - (a+1)$ is divisible by p .

By using the binomial theorem, we find that

$$\begin{aligned} (a+1)^p - (a+1) &= a^p + pa^{p-1} + C_2^p(a^{p-2}) + C_3^p(a^{p-3}) + \dots + pa + 1 - a - 1 \\ &= (a^p - a) + pa^{p-1} + C_2^p(a^{p-2}) + C_3^p(a^{p-3}) + \dots + pa. \end{aligned}$$

But all the binomial coefficients

$$C_k^p = \frac{p(p-1)(p-2)\dots(p-k+1)}{k(k-1)(k-2)\dots 4 \cdot 3 \cdot 2 \cdot 1}$$

cont. bottom p. 17



Fermat's Last Theorem

by David Zagorski

Pierre Fermat was a 17th century lawyer whose hobby was studying mathematics. Acknowledged as one of the premier mathematicians in Europe of this period, Fermat is primarily remembered for his work in number theory. His most famous result is a conjecture which has come to be known as Fermat's Last Theorem.

The fascinating story of Fermat's Last Theorem began in the 17th century B.C. when the Babylonians discovered Pythagorean triples, i.e., positive integers satisfying the equation $x^2 + y^2 = z^2$. In the 1630's, Fermat was reading a Latin translation of Diophantus' "Arithmetic" when he came across a problem challenging the reader to generate Pythagorean triples. Fermat responded by writing in the margin next to the problem, "On the other hand, it is impossible for a cube to be written as the sum of two cubes, or a fourth power to be written as the sum of two fourth powers, or, in general, for any number which is a power greater than the second to be written as the sum of two like powers. I have a truly marvelous demonstration of this proposition which this margin is too narrow to contain."

Unfortunately, these are the only written records which Fermat left dealing with this conjecture, which is usually stated as:

"For any integer $n > 2$, it is impossible to find positive integers satisfying $x^n + y^n = z^n$."

If Fermat did indeed prove his conjecture, it was a truly remarkable feat, for it has not been duplicated in the 300 years since his death. During this era numerous mathematicians dedicated their lives to proving Fermat's conjecture, but were unsuccessful in doing so. To date, many incorrect "proofs" have been offered and published in various journals.

Despite the absence of a general proof of Fermat's conjecture, a proof exists for all exponents well into the thousands. A method commonly used to prove specific cases of Fermat's assertion involves infinite descent (see p.18). This technique was employed by Euler to prove the case $n=3$ and by Dirichlet and Legendre to prove the case $n=5$.

I present here a proof of the following special case:

Theorem: If p and $2p+1$ are both primes and $2p+1 \nmid xyz$ then there are no solutions to $x^p + y^p = z^p$.

Proof: From Fermat's Little Theorem (see p.16), $x^{2p} \equiv 1 \pmod{2p+1}$. Thus $x^p \equiv \pm 1 \pmod{2p+1}$. Similarly, $y^p \equiv \pm 1 \pmod{2p+1}$ and $z^p \equiv \pm 1 \pmod{2p+1}$. Considering $x^p + y^p = z^p \pmod{2p+1}$ yields $(\pm 1) + (\pm 1) = (\pm 1)$. Such an arrangement is impossible.

Q.E.D.

are divisible by the prime number p , since C_k^p is an integer and since the numerator of the above expression contains the factor p whereas the denominator does not contain this factor. Now, by the induction hypothesis, $a^p - a$ is also divisible by p , so it follows that $(a+1)^p - (a+1)$ is divisible by p .

Q.E.D.

Infinite Descent

by Larry Buxbaum

Pierre de Fermat did much work in the mathematics of integers, known as number theory. Besides Fermat's Little Theorem and the long-standing unsolved Last Theorem, Fermat also left us his ingenious method of infinite descent. This form of proof was Fermat's favorite, and he used it to solve many problems.

Infinite descent is a method of proving the nonexistence of positive integers $a, b, c, \dots$ that satisfy some relation. We first assume such integers exist and show that consequently, positive integers $a_1, b_1, c_1, \dots$ also exist which satisfy the same relation with $a > a_1, b > b_1, c > c_1, \dots$. By repetition, we can show that positive integers $a_2, b_2, c_2, \dots$ also satisfy the original conditions with $a_1 > a_2, b_1 > b_2, c_1 > c_2, \dots$. Hence we have infinitely many sets of positive integers of the form $a_i, b_i, c_i, \dots$ that satisfy the conditions of the original relation. However, the number of positive integers less than a is finite. This contradicts what we have found, and therefore, the original hypothesis that positive integers $a, b, c, \dots$ exist is incorrect.

Two problems of this sort will be examined in this article. The first is to prove that $\sqrt{3}$ is irrational. We assume the contrary, that there are positive integers a and b such that $\sqrt{3} = a/b$.

If this is indeed true, then from the equality $2 + \sqrt{3} = 1/(2 - \sqrt{3})$, it follows that

$$2 + \frac{a}{b} = \frac{1}{2 - \frac{a}{b}}.$$

Solving for a/b , we obtain

$$\frac{a}{b} = \frac{2a - 3b}{2b - a}.$$

Suppose $\frac{2a - 3b}{2b - a} = a_1/b_1$. Since $a/b = \sqrt{3}$, $1 < a/b < 2$. Multiplying this inequality by b yields $b < a < 2b$. Since $2b > a$, clearly $3b > a$. Therefore, $a + 3b > 2a$, which implies $a > 2a - 3b = a_1$. Furthermore, adding $b - a$ to both sides of the inequality $a > b$ yields $b > 2b - a = b_1$. Thus, we have obtained the following results:

$$a_1 < a; b_1 < b; a_1/b_1 = a/b.$$

This process can be repeated as often as we please. But there are a finite number of positive integers less than a . Therefore, our original assumption that $\sqrt{3}$ is expressible in the form a/b , where a and b are positive integers, must be incorrect.

The second problem that can be solved using infinite descent requires some knowledge of divisibility and modular arithmetic. By definition $a \equiv b \pmod{m}$ (read " a is congruent to b modulo m ") if m is a factor of $a - b$, where a, b and m are integers and m is positive. For example, $13 \equiv 5 \pmod{4}$ since 4 is a factor of $13 - 5$.

We are seeking positive integral solutions to the equation $15x^2 - 4y^2 = 3z^2$. Since there are three variables and only one equation, we call this type of equation an indeterminate one (it has an infinite number of real solutions). We wish to prove that there are no ordered triples (x, y, z) where x, y and z are positive integers which satisfy this equation.

Consider our original equation mod 3. Since both $15x^2$ and 3^z are multiples of 3,

$$-4y^2 \equiv 0 \pmod{3}.$$

Since -4 is not a multiple of 3, $y^2 \equiv 0 \pmod{3}$, which implies that

$$y \equiv 0 \pmod{3}.$$

Therefore, y is a multiple of 3. Let $y=3y_1$. Substituting this value of y into the original equation yields $15x^2-36y_1^2 = 3^z$, or

$$(i) \quad 5x^2-12y_1^2 = 3^{z-1}.$$

Considering (i) mod 3, we obtain $5x^2 \equiv 0 \pmod{3}$, which as before implies

$$x \equiv 0 \pmod{3}.$$

Hence x is also a multiple of 3. Let $x=3x_1$. Substituting this value of x into (i) yields $45x_1^2-12y_1^2=3^{z-1}$, or

$$15x_1^2-4y_1^2 = 3^{z-2}.$$

Let $z_1=z-2$. Thus $15x_1^2-4y_1^2=3^{z_1}$, (note the similarity between this equation and the one we began with) and if (x,y,z) is a solution of this equation, then there must also be a solution (x_1, y_1, z_1) , with $x_1 < x$, $y_1 < y$, and $z_1 < z$.

In this problem, there is one modification in the process of infinite descent, because we must stop when $z=0$, since $3^0=1$ is not a multiple of 3. Since each step involves diminishing z by 2, when z is even the equation can be reduced to

$$15x_1^2-4y_1^2 = 3^0 = 1, \text{ but when } z \text{ is odd the equation can only}$$

be reduced to

$$(ii) \quad 15x_j^2-4y_j^2 = 3.$$

We will consider the two cases z even and z odd separately. First, if z is even then $15x_1^2-4y_1^2 = 1$. Considering this equation mod 3 we obtain $-4y_1^2 \equiv 1 \pmod{3}$. Therefore, $y_1^2 \equiv 2 \pmod{3}$, which is impossible because all numbers are of the form $3k$, $3k+1$, or $3k-1$, and $(3k)^2 \equiv 0 \pmod{3}$ and $(3k\pm 1)^2 \equiv 1 \pmod{3}$. This means that z must be odd.

For odd z , $15x_j^2-4y_j^2 = 3$ implies $-4y_j^2 \equiv 0 \pmod{3}$. Therefore, $y_j^2 \equiv 0 \pmod{3}$ and $y_j \equiv 0 \pmod{3}$. Let $y_j=3y_{j+1}$. Substituting this into (ii) yields $15x_j^2-36y_{j+1}^2 = 3$. Dividing by 3 yields $5x_j^2-12y_{j+1}^2 = 1$, and proceeding as we did before we see that $x_j^2 \equiv 2 \pmod{3}$. Again this is impossible. We must conclude that

the original assumption that there are positive integers (x,y,z) that satisfy the equation $15x^2-4y^2=3^z$ is incorrect.

The techniques illustrated above can be used to solve problems similar to the ones given. Can you show that $\sqrt{63}$ is irrational? Can you show the non-existence of positive integers (x,y,z) such that $10x^2+17y^2=5^z$? If so, then you have mastered the skills of infinite descent.

Regents Refresher

Quadratic Formula: If $ax^2+bx+c=0$, then $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$, where $d=b^2-4ac$ is the discriminant.

If $d>0$, then the roots are real and distinct.

If $d=0$, then the roots are real and equal.

If $d<0$, then the roots are complex and are conjugates of each other.

Equations of Plane Curves:

Line: $y = mx + b$, where m =slope and b =y intercept

OR

$y-y_1 = m(x-x_1)$, where (x_1, y_1) is a point on the line and m =slope

Circle: $(x-h)^2+(y-k)^2=r^2$, where (h,k) is the center and r =radius

Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a and b are the major and minor axes

Parabola: $(x-h)^2 = 2p(y-k)$, where (h,k) is the vertex and p is the distance from vertex to directrix

Hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where a and b are the major and minor axes

Logarithms

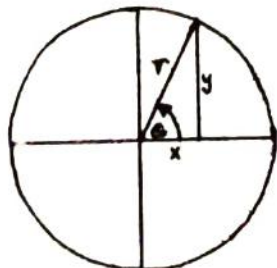
$\log_B N = E$ is equivalent to $B^E = N$.

$\log_c 1 = 0$ $\log_c c = 1$ $\log_c c^b = b$ $\log_c a^b = b \log_c a$

$\log_c^b a = (\log_c a)/b$ $\log_c ab = \log_c a + \log_c b$ $\log_c (a/b) = \log_c a - \log_c b$

$$\log_b a = \frac{\log_c a}{\log_c b}$$

Trigonometry



$$r = \sqrt{x^2+y^2}$$

SOH-CAH-TOA

$$\sin \theta = y/r$$

$$\cos \theta = x/r$$

$$\tan \theta = y/x$$

$$\csc \theta = r/y$$

$$\sec \theta = r/x$$

$$\cot \theta = x/y$$

$$\sin \theta = \cos(90-\theta)$$

$$\tan \theta = \cot(90-\theta)$$

$$\sec \theta = \csc(90-\theta)$$

Amplitude of $n \sin \theta = n$

Period of $n \sin \theta = 360/r$

$$\sin^2 \theta + \cos^2 \theta = 1; \quad 1 + \tan^2 \theta = \sec^2 \theta;$$

$$\cot^2 \theta + 1 = \csc^2 \theta.$$

$$\sin 2A = 2\sin A \cos A \quad \cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$$\tan 2A = 2\tan A / (1 - \tan^2 A)$$

$$\sin \frac{1}{2}A = \pm \sqrt{\frac{1 - \cos A}{2}} \quad \cos \frac{1}{2}A = \pm \sqrt{\frac{1 + \cos A}{2}} \quad \tan \frac{1}{2}A = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}} = \csc A - \cot A$$

$$\begin{aligned} \sin(A+B) &= \sin A \cos B + \cos A \sin B & \cos(A+B) &= \cos A \cos B - \sin A \sin B \\ \sin(A-B) &= \sin A \cos B - \cos A \sin B & \cos(A-B) &= \cos A \cos B + \sin A \sin B \\ \tan(A+B) &= (\tan A + \tan B) / (1 - \tan A \tan B) \\ \tan(A-B) &= (\tan A - \tan B) / (1 + \tan A \tan B) \end{aligned}$$

Geometry

(A=Angle, S=Side)

Two triangles are congruent if you can prove SSS, SAA or ASA.

Two triangles are similar if you can prove AA.

Two right triangles are congruent if you can prove that their hypotenuses are equal and that one of their legs are equal.

The sum of the angles in a triangle is 180 degrees.

In a right triangle with legs a and b and hypotenuse c , $a^2 + b^2 = c^2$.

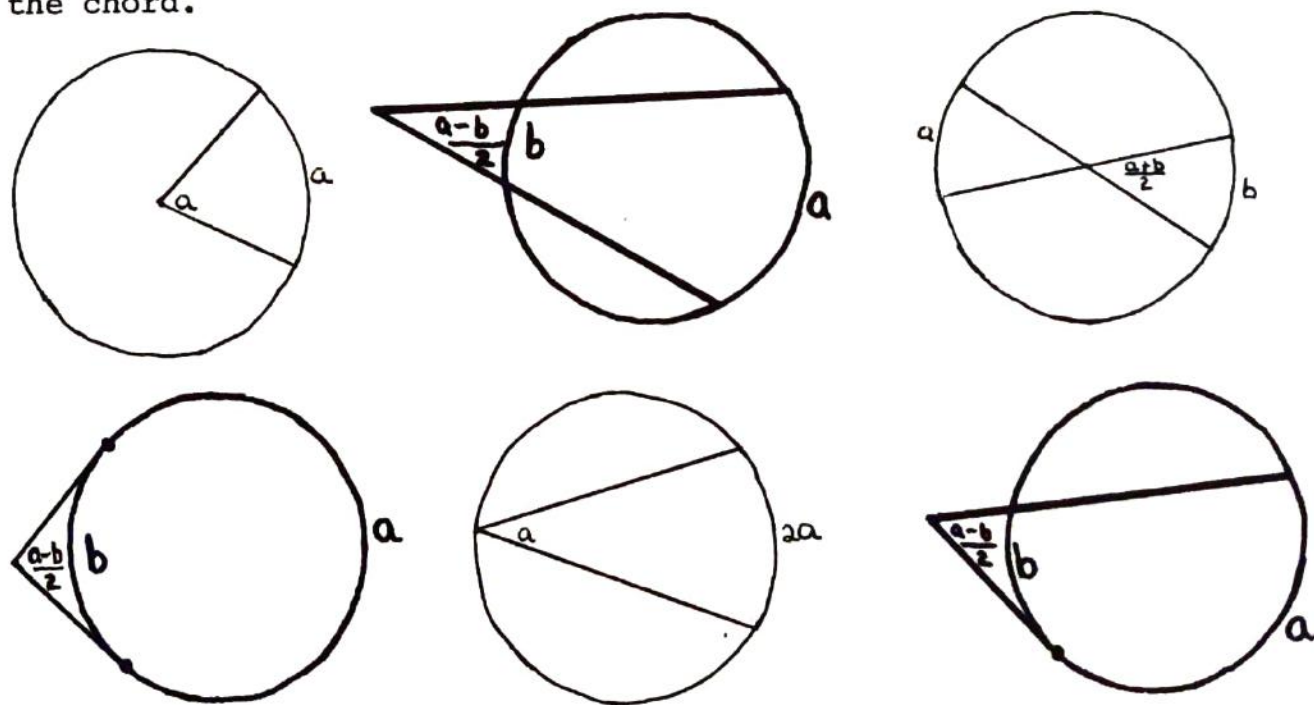
The area of a triangle is $\frac{1}{2}bh$ (h =altitude to side of length b).

The area of a trapezoid is $\frac{1}{2}(b+B)h$

The area of a parallelogram is bh .

The area of a regular n -gon is $\frac{1}{2}ap$ (a =apothem, p =perimeter).

In a circle, the diameter that is perpendicular to a chord bisects the chord.



Translation:

A Mathematical Limerick

$$\frac{12+144+20+3\sqrt{4}}{7} + 5 \cdot 11 = 9^2 + 0.$$

A dozen, a gross and a score
Plus 3 times the square root of 4,
Divided by 7
Plus 5 times 11,
Equals 9 squared and not a bit more.

Puzzle by Jay Sipelstein

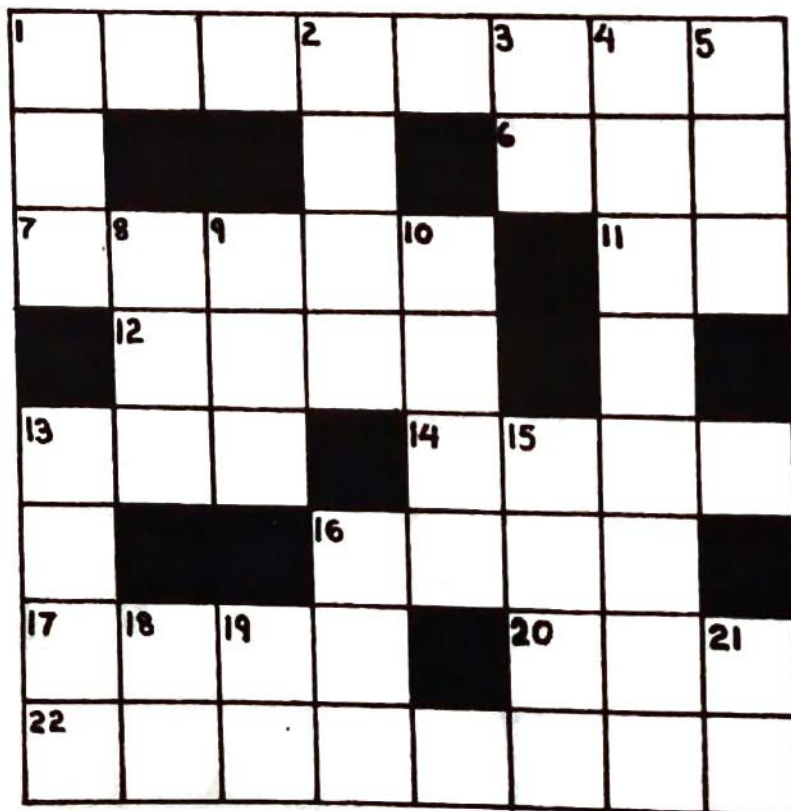
(round all answers to the nearest integer)

Across

1. $e \cdot 10^7$
6. Express 7^2 as an integer in base 5.
7. (1 down)²
11. Prime
12. $3^{2^0} + 3^{2^1} + 3^{2^2} + 3^{2^3}$
13. $6(81 \cdot 8 - (5 \cdot 62 + 67)) - 6!$
14. (palindrome) $\cdot 10 + 9$
16. $x+y=70$
 $7x=3y$
Find $100x+y$.
17. $a+b=135$
 $28a=17b$
Find $100a+b$.
20. Prime
22. $15!/8! + 70$

Down

1. An integer of the form x^x
2. $a+b=cd+2$; find a,b,c,d
3. $(\sqrt[3]{3})^8$
4. $\sqrt[3]{5!}$
5. Square
8. $7!/9$
9. $n=1111_2+1111_3+1111_4+1111_5+1111_6+11_{10}$. (X_b means that the number X is in base b)
Find n_{10} .
10. $\sqrt{42 \times 106}$
13. $a+b+c+d=24$; $a-c=4$. Find a,b,c,d
15. The least perfect square ending in the same three digits
16. (3 down) $\cdot 3$
18. $\log_2 8^4$
19. Even
21. $\log \log 10^{10^{10}}$ (both logs are base 10)



Math Society

This year the Math Society has undergone a successful evolution to appeal to all Stuyvesant students. As in the past, Math Society has presented numerous distinguished speakers who delivered informal lectures dealing with a wide range of topics in mathematics. Even though the topics discussed were advanced and are not usually taught in any math course offered at Stuyvesant, they are understandable to most Stuyvesant students. Math Society co-presidents James Balter and David Zagorski, vice-presidents Jeff Saltiel and Jerry Schreiberstein, and faculty advisor David Greenfield have brought Richard Churchill, Harry Hochstadt, David Kelly (the director of the Hampshire College Summer Studies in Mathematics program), Joseph Malkevitch, Harry Ruderman, Harold Shapiro and William Wernick to speak before us thus far. The topics discussed this year have included game theory, geometric constructions, non-Euclidean geometry, random walk problems, topology, and unsolved problems in number theory.

In addition, on February 18, 1982, Math Society held its first annual Rubik's Cube-athon. Close to 70 Stuyvesant students and one teacher competed in this event. After the last cube was rotated, the following six people, in order of finish, emerged victorious: James Balter, Carl Walker, Tzea-Ning Chen, David Read (who notched the fastest time of 29 seconds in a preliminary round), Mona Kim and Jimmy Lin. They received beautiful certificates for their accomplishments.

If the response is sufficiently great, we will have a math SAT/Achievement Test review after school on some day before the June testing date, as well as a talk given by one or more of the Westinghouse Science Talent Search winners about their projects and about guidelines that entrants in this prestigious contest should follow.

Math Society meets after 8th period on selected Tuesdays and Thursdays in room 215. Details of upcoming events can be obtained in Mr. Marcantonio's office. Stop by and join us! Service credit will be given to those who attend meetings regularly.

COMPLIMENTS

OF

THE STUYVESANT HIGH SCHOOL

PARENTS' ASSOCIATION

What's a Fractal?!

by Jeff Rosenfeld

Fractals are very unusual variations from the standard geometry of Euclid. Imagine curves that have no tangents, infinitely long paths that enclose finite areas, landscape surfaces that wiggle part way into the next dimension —

What is probably the most basic fractal is the triadic Koch curve, or more commonly, the "snowflake" curve (see Figure 1). It has a dimension of $\log 4 / \log 3$ (approx. 1.26186). This curve can be constructed in the following way: Begin with an equilateral triangle, and construct new equilateral triangles outside the original one using the middle third of each side of the original triangle as a base. Continuing this process ad infinitum yields the triadic Koch curve.



Figure 1

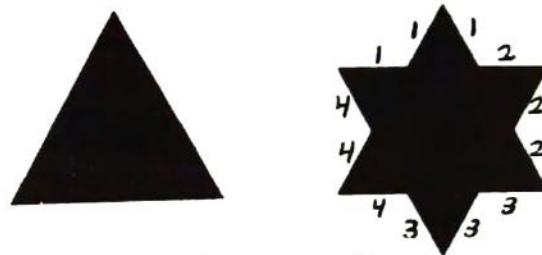


Figure 2

Koch curves have some remarkable properties. Firstly, every point on such a curve is a "corner". Consequently, although the curve is continuous, it has no tangents. Secondly, the curve violates the postulate of Euclidean geometry which states that any region with a finite area has a finite perimeter. While the area bounded by the curve is $\sqrt{3}$, the perimeter, which increases by a factor of $4/3$ each time the construction described above is repeated, is clearly infinite. Third, Koch curves are self-similar, that is, they appear the same at any scale of measurement.

Earlier, I mentioned that the dimension of the triadic Koch curve is $\log 4 / \log 3$. The reason why it is such a strange number is that the dimension of a fractal is calculated from the formula

$$D = \log N / \log(1/r), \quad (1)$$

where D stands for the dimension and r is the relative size of each section when the figure is divided into N congruent sections.* For the Koch curve, observe from figure 2 that the dimension is calculated easily if the curve is partitioned into 4 equal parts because the perimeter of each is $1/3$ of the original perimeter. Thus, substituting $N=4$ and $r=1/3$ into (1) yields the dimension of the Koch curve.

By now you know more about fractals than perhaps even Professor Riemann, so I will have to leave you with this — fractals have wide applications in both practical and theoretical mathematics, and some time in the future, they might even become part of the school geometry curriculum.

* The method by which N and r are calculated is found in Fractals, Form, Chance and Dimension by Benoit B. Mandelbrot.

Westinghouse Tips

So you want to do a Westinghouse project. The first and most important step you must take is to select a topic. This may sound simple, but some winners (the editors included) will concede that finding their topics gave them more headaches than doing their research. Bear in mind that your goal should be to restrict your area of study to a problem which satisfies these criteria: 1) It should interest you. If it doesn't, choose another topic, because you will find that your work will get monotonous after a while. 2) Make sure that the topic is not "over your head" — be honest with yourself about this. 3) You should have a specific question in mind (which preferably has not been resolved by anyone else) that you hope to answer. In recent years, many winning projects have dealt with specific questions about number theoretic functions (such as those of Euler), and Diophantine equations.

Once you have chosen your topic, you should check to see what has already been written about it. The mathematics libraries of universities such as Columbia and N.Y.U. are extremely helpful because they have not only an extensive collection of books but also numerous mathematical journals. Although it is only in rare instances that you will be permitted to borrow materials from these libraries, both have copying machines which you should use liberally.

Now you are ready to begin your research. Of course, the more time you allot for this, the better off you are likely to be. While many winning projects have been done over a period of two or three years, just as many have been the result of a month or two of hard work.

There are two points I cannot stress enough. First, don't be discouraged by apparent failure. You will probably find yourself experiencing frustration to some degree about as often as there is a traffic jam in New York. However, these feelings do pass. Second, don't try to judge the significance of your results. Even though you may feel that your work is pointless and has no value, enter it in any case. Very few winning projects have resolved world crises.

If all has gone according to plan, you will have completed the bulk of your research by November. By that time you should seriously think about writing your project report. Very often, while you are writing, you will notice magnificent things which have been sitting before your eyes all along which you hadn't seen previously. Unfortunately, you will also discover flaws and missing steps in proofs, which you will have time to fix. At this time, you should also begin working on the Personal Data Blank, a questionnaire which is at least as important as the project itself. The Data Blank solicits specific examples from both you and one of your teachers which illustrate your scientific attitude, curiosity, inventiveness, initiative and work habits. You should spend a great deal of time compiling a list of activities in each category and preparing detailed but concise responses for each question.

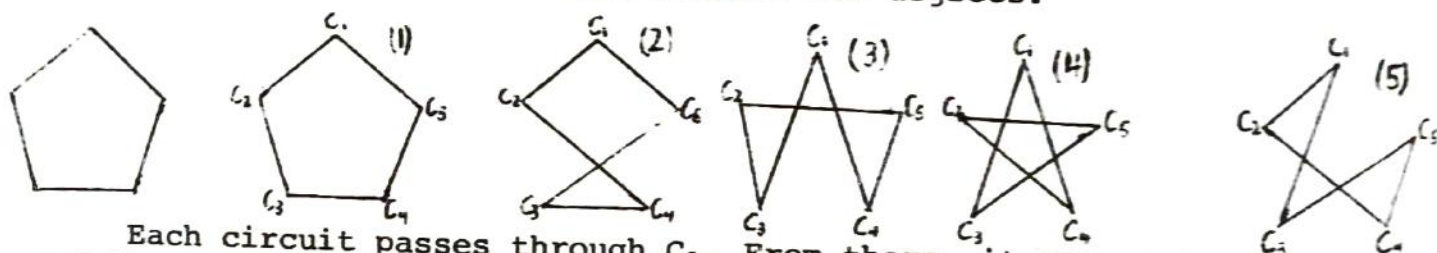
Hopefully, all will go smoothly, and by midnight of December 15th (the due date), your materials will be in Washington D.C. From that point on, you can only wait anxiously until the winners are selected. Work hard and Good Luck!

Wilson's Theorem

by Noam Elkies

Wilson's Theorem states that, if p is a prime number (that is, a whole number with no divisors other than 1 or itself, such as 5 or 1,234,567,891) then the product of all natural numbers from 1 to $(p-1)$ is 1 less than a multiple of p . For example, when $p=5$, we have $1 \cdot 2 \cdot 3 \cdot 4 = 5 \cdot 5 - 1$. The interested (and insane) reader can verify that the same result holds for $p=1,234,567,891$. The proof which is most commonly offered for this surprising result depends on the notion of the inverse of a number modulo p . I will now present a simpler geometric proof.¹

Consider a regular p -gon. Its corners, which I'll call $C_1, C_2, C_3, \dots, C_p$, are connected by a closed circuit consisting of p line segments. We now ask: in how many ways can the task of connecting the p corners be accomplished? The circuit can intersect itself as often as we want, but two circuits are not "different" if one can be rotated so that it coincides with the other. The diagram below shows the four possible circuits for $p=5$; the fifth circuit is just the third circuit rotated 144 degrees.



Each circuit passes through C_1 . From there, it can go to any of the remaining $(p-1)$ corners. After it reaches the second corner it can choose among the $(p-2)$ untouched corners for its next stop. Hence we have $(p-1)(p-2)$ ways to draw the first 2 lines. Similarly, there are $(p-1)(p-2)(p-3)$ ways to draw the first 3 lines, and by continuing the process, we find that there are $(p-1)(p-2) \dots (2)(1)$ ways to draw the entire circuit. This last product is usually abbreviated as $(p-1)!$ — read "pea minus one factorial".

We're not done yet, since we counted each circuit more than once. In fact, we counted many of the circuits p times, once for each rotation. The only exceptions are circuits like (1) and (4) above which are symmetrical — that is, unchanged by some rotations — and have fewer than p different forms. Assume that some circuit is unchanged by rotating C_1 to, say, C_{n+1} . By repeating the rotation we bring C_{n+1} to C_{2n+1} , C_{2n+1} to C_{3n+1} , Whenever the subscript exceeds p , we will reduce it by p as many times as necessary to make it an integer between 1 and p . For example, when $p=5$, C_8 , C_{13} , C_{18} , C_{23} , etc. all refer to the same corner, i.e. C_3 . Since p is prime, by the Chinese Remainder Theorem² every possible rotation is thus effected. Hence our circuit must be symmetrical under all the rotations. The only such circuits are the regular p -gon and the "stars" such as (4) above — that is, circuits of the form $C_1, C_{k+1}, C_{2k+1}, C_{3k+1}, \dots, C_{(p-1)k+1}, C_1$.

¹"I don't know why this guy Fermat is so great. His last theorem wasn't even a theorem, and his little theorem was generalized by Gauss."

In such circuits, k can have the $(p-1)$ values $1, 2, 3, \dots, (p-1)$. Therefore, there are $(p-1)$ such circuits. The rest of them, which are $(p-1)!(p-1)$ in number, are really identical p -tuplets, and thus account for only $[(p-1)!(p-1)]/p$ different circuits. Adding the $(p-1)$ stars yields, after some simple manipulation, a total of $\{[(p-1)!+1]/p\}+p-2$ circuits. Finally, since a closed circuit can be traversed in either of two directions, we have counted each circuit twice. Consequently, we must divide the last figure by 2 to obtain the formula:

$$\text{Number of circuits} = \frac{1}{2} \left(\frac{(p-1)!+1}{p} \right) + p - 2 .$$

Note that for $p=5$, this formula gives 4 circuits, which we found earlier to be correct.³ Now, the number of circuits must be a whole number; thus, $[(p-1)!+1]/p$ is an integer for all primes p , completing the proof.

-
- (1) This proof was pointed out by Professor Andrew Liu in the summer of 1981.
 - (2) The Chinese Remainder Theorem states: If n and p have no common factors, then for every integer r one can find another integer a such that $a \cdot n$ is r more than a multiple of p . Here this means that any rotation by $r(360/p)^\circ$ can be expressed as the equivalent of a certain number a of $n(360/p)^\circ$ rotations.
 - (3) If $p=7$, the number of circuits is $\frac{1}{2} \left(\frac{6!+1}{7} + 7 - 2 \right) = 54$. See if you can find at least 25 of them.
-

Mathematical Miscellany

1. Raiders of the Lost Arc: An adventure story with a somewhat circular story-line.
2. From Here to Eternity: Here, Here+1, Here+2, ... (monotonous after a while).
3. Vector/Vectoria: Is it $\vec{AB}$? Or is it $-\vec{BA}$? Or is it $-(\vec{-AB})$?...
4. Apoc-ellipse Now: The Oval Office takes action abroad.
5. The 39 Steps: An unusually long geometry proof.
6. The Old Man and the c: Einstein in his later years.
7. For your i's Only: Spy thriller with a complex plot.
8. Roots: Alex Haley traces the origin of $\sqrt{2}, \sqrt{3}, \dots$
9. Real People: Guests are (or is that R?): a $\forall$ baker, and a composer of a quartet in e.

Vector Proofs

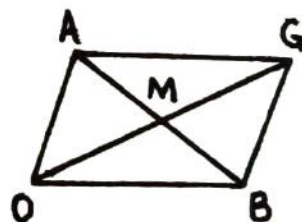
by David Zuckerman

Vector algebra is a very useful mathematical tool since many geometric problems can be solved more easily by using vectors than by using conventional geometric methods. Vectors have been used for over 2,300 years, and even Euclid employed them in his proofs. We will develop some elementary concepts of vector algebra and then give an example of a geometric theorem that can be proven through vector analysis.

A vector is a directed line segment (it has both direction and magnitude). For example, $\overrightarrow{AB}$ represents the directed line segment from A to B. In the proofs that follow, we will choose an arbitrary point O, and represent all vectors $\overrightarrow{AB}$ in terms of $\overrightarrow{OA}$ and $\overrightarrow{OB}$. To be concise, we let $\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{OB}=\mathbf{b}$, and in general, $\overrightarrow{OX}=\mathbf{x}$. Using vector addition from physics, $\overrightarrow{AB}=\overrightarrow{AO}+\overrightarrow{OB}=-\overrightarrow{OA}+\overrightarrow{OB}=-\mathbf{a}+\mathbf{b}=\mathbf{b}-\mathbf{a}$.

Theorem: The midpoint M of line segment $\overline{AB}$ corresponds to $\mathbf{m}=(\mathbf{a}+\mathbf{b})/2$.

Proof: We may construct parallelogram OAGB, where $\mathbf{g}=\overrightarrow{OG}=\overrightarrow{OA}+\overrightarrow{AG}=\overrightarrow{OA}+\overrightarrow{OB}=\mathbf{a}+\mathbf{b}$. Since the diagonals of a parallelogram bisect each other, if $\mathbf{m}=(\mathbf{a}+\mathbf{b})/2$, M will lie on $\overline{AB}$. As $\overrightarrow{AM}=\overrightarrow{MB}$, M is the midpoint of $\overline{AB}$.



Q.E.D.

In general, it can be shown that the centroid P of an n-gon $A_1A_2\dots A_n$ is determined by $\mathbf{p}=(\mathbf{a}_1+\mathbf{a}_2+\dots+\mathbf{a}_n)/n$.

Theorem: Point C will lie on line segment $\overline{AB}$ if and only if there exists a real λ , $0\leq\lambda\leq 1$, such that

$$\mathbf{c} = \lambda \cdot \mathbf{a} + (1-\lambda) \cdot \mathbf{b}.$$

If this holds, $\overline{AC}:\overline{CB} = 1-\lambda:\lambda$.

Proof: To show the necessity of the condition, we suppose that C lies on $\overline{AB}$. We construct $\overline{CD} \parallel \overline{OB}$ and $\overline{DE} \parallel \overline{AB}$, with D and E lying on $\overline{OA}$ and $\overline{OB}$, respectively. Let $\overrightarrow{OD}=\lambda \cdot \overrightarrow{OA}=\lambda \cdot \mathbf{a}$. Since $\overrightarrow{OD} \leq \overrightarrow{OA}$ and $\overrightarrow{OD}$ has the same direction as $\overrightarrow{OA}$, $0 \leq \lambda \leq 1$. As triangle ODE is similar to triangle OAB, $\overrightarrow{OE}=\lambda \cdot \overrightarrow{OB}=\lambda \cdot \mathbf{b}$. $\overrightarrow{EB}=\overrightarrow{OB}-\overrightarrow{OE}=\mathbf{b}-\lambda \cdot \mathbf{b}=(1-\lambda) \cdot \mathbf{b}$. Since CDEB is a parallelogram, $\overrightarrow{DC}=\overrightarrow{EB}=(1-\lambda) \cdot \mathbf{b}$. Therefore, $\mathbf{c}=\overrightarrow{OC}=\overrightarrow{OD}+\overrightarrow{DC}=\lambda \cdot \mathbf{a}+(1-\lambda) \cdot \mathbf{b}$.

To show the sufficiency of the condition, we assume that $\mathbf{c}=\lambda \cdot \mathbf{a}+(1-\lambda) \cdot \mathbf{b}$ for some real λ , $0\leq\lambda\leq 1$. Then we may construct parallelogram CDEB such that $\overrightarrow{OD}=\lambda \cdot \mathbf{a}$. Since $\mathbf{c}=\lambda \cdot \mathbf{a}+(1-\lambda) \cdot \mathbf{b}$, $\overrightarrow{DC}=\overrightarrow{OC}-\overrightarrow{OD}=\lambda \cdot \mathbf{a}+(1-\lambda) \cdot \mathbf{b}-\lambda \cdot \mathbf{a}=(1-\lambda) \cdot \mathbf{b}$. As demonstrated above, this occurs when C lies on $\overline{AB}$.

To prove that $\overline{AC}:\overline{CB} = 1-\lambda:\lambda$, we again use the similar triangles ODE and OAB. $\overrightarrow{OD}=\lambda \cdot \mathbf{a}$ implies that $\overrightarrow{DE}=\lambda \cdot \overrightarrow{AB}$. Because CDEB is a parallelogram, $\overrightarrow{CB}=\overrightarrow{DE}=\lambda \cdot \overrightarrow{AB}$ and $\overrightarrow{AC}=\overrightarrow{AB}-\overrightarrow{CB}=\overrightarrow{AB}-\lambda \cdot \overrightarrow{AB}=(1-\lambda) \cdot \overrightarrow{AB}$. Therefore, $\overline{AC}:\overline{CB} = 1-\lambda:\lambda$.

Q.E.D.

Now we are ready for a relatively simple vector proof of a geometric theorem.

Theorem: The medians of triangle RST coincide at the centroid, and divide each other in a 1:2 ratio.

Proof: Let D, E and F be the midpoints of $\overline{RS}$, $\overline{RT}$ and $\overline{ST}$, respectively. Then

$$(1) d = (r+s)/2$$

$$(2) e = (r+t)/2$$

$$\text{and } (3) f = (s+t)/2.$$

We already know that the centroid P can be represented by

$$(4) p = (r+s+t)/3.$$

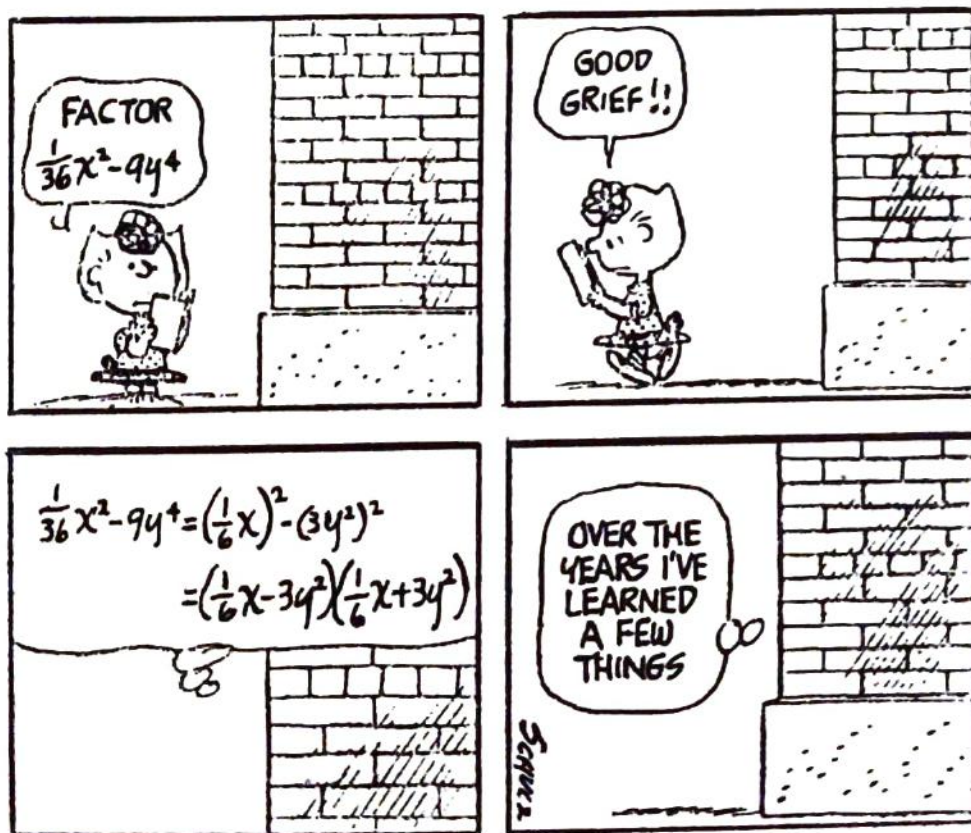
From (1), we may substitute $2d$ for $(r+s)$ in (4) to obtain

$$p = (2d+t)/3 \text{ or}$$

$$(5) p = 2d/3 + t/3.$$

Since $2/3 + 1/3 = 1$, we know from our second theorem that P lies on $\overline{DT}$. (In this case, $\lambda = 2/3$.) In addition, P divides $\overline{DT}$ in the ratio $\overline{DP}:\overline{PT} = 1:2$. Using the same procedure (substituting $2e$ for $r+t$ and $2f$ for $s+t$ in (4)), we can show that P lies on $\overline{ES}$ and $\overline{FR}$, and divides them in ratios of 1:2.

Q.E.D.



Special Pythagorean Triples *by Kenneth Baron*

Over 2,500 years ago, Pythagoras proved that the sum of the squares of the lengths of the two legs of a right triangle is equal to the square of the length of the hypotenuse. In other words, if a and b are the lengths of the legs and c is the length of the hypotenuse, then

$$a^2 + b^2 = c^2. \quad (1)$$

A proof of this theorem can be found in any elementary geometry text.

When a, b and c are positive integers, (a, b, c) is called a Pythagorean triple. In this article, we will solve two problems of interest: we will first determine all triples in which $c-a=1$ or $c-b=1$, and second, we will generate all triples containing a given number.

We shall make use of a formula that generates all relatively prime triples (i.e. ones in which a, b and c have no factors in common):

$$a = m^2 - n^2; b = 2mn; c = m^2 + n^2 \quad (2)$$

where m and n are relatively prime positive integers, $m > n$.

I: Note that if $c-a=1$ or $c-b=1$, then a, b and c can have no factors in common. Thus formula (2) will generate all triples such that $c-a=1$ or $c-b=1$. We now must find for what values of m and $n, c-a=1$ or $c-b=1$.

We first consider the case $c-a=1$.

$$(m^2 + n^2) - (m^2 - n^2) = 1;$$

$$2n^2 = 1;$$

$$n = \sqrt{2}/2.$$

But n must be integral, so the case $c-a=1$ has no solutions.

If $c-b=1$, then

$$m^2 + n^2 - 2mn = 1;$$

$$(m-n)^2 = 1;$$

$$(m-n) = 1.$$

In other words, setting m equal to $n+1$ yields all triples in which $c-b=1$. Substituting $m=n+1$ into (2) gives us

$$a = 2n+1; b = 2n^2+2n; c = 2n^2+2n+1.$$

For example, $n=8$ produces the triple $(17, 144, 145)$.

II: In order to generate all triples containing an integer x , we consider the divisors of x , $d_1, d_2, d_3, \dots, d_k = x$ where $x > d_{k-1} > \dots > d_3 > d_2 > d_1 > 1$.

We will now show that all triples containing x can be expressed as (Fa, Fb, Fc) , or $F(a, b, c)$ where (a, b, c) is a relatively prime triple containing a d_i .

Proof: Suppose we have a triple (a', b', c') , where a', b' or c' equals x . If the triple is relatively prime, we are done, because we have a relatively prime triple containing a divisor of x , namely x itself. In this case, $F=1$.

If the triple is not relatively prime, then two of the members of the triple must have a factor in common. Let $a'=aF_1$ and $b'=bF_1$. Then

$$(aF_1)^2 + (bF_1)^2 = c'^2.$$

But the left side of the equation is a multiple of F_1 , so the right side must be a multiple of F_1 as well. Thus, $c'=F_1c''$, and we have the triple (aF_1, bF_1, cF_1) , which we can write as $F_1(a'', b'', c'')$. Since one of a' , b' or c' was x , then a'' , b'' or c'' must be x/F_1 , or a divisor of x . We can repeat this process to show that either (a'', b'', c'') is relatively prime or that we can write (a'', b'', c'') as $F_2(a''', b''', c''')$, where one of a''' , b''' , and c''' is a divisor of x . Thus, we can write (a', b', c') as $F_1F_2 \dots (a, b, c) = F(a, b, c)$ where (a, b, c) is a relatively prime triple and one of a, b and c is a divisor of x . Q.E.D.

Notice the similarity of this technique to that of infinite descent (see p. 18).

We are now almost done. We know that every triple containing x can be generated from a relatively prime triple containing a divisor of x , and, using (2), we can generate every relatively prime triple. In other words, we must find all relatively prime triples containing a divisor of x . This can be accomplished by setting each d_i equal to m^2-n^2 , $2mn$ and m^2+n^2 , in turn.

This time-consuming task can be shortened somewhat by considering (2) modulo 4. We can see that m and n must be of opposite parities (i.e. one is odd and one is even), or else the triple will not be relatively prime (all three members of the triple will be even). Since any odd number squared is congruent to 1 modulo 4,

$$m^2 - n^2 \equiv \pm 1 \pmod{4},$$

$$2mn \equiv 0 \pmod{4} \quad \text{and}$$

$$m^2 + n^2 \equiv 1 \pmod{4}.$$

Thus, we need not consider divisors that are congruent to 2 (mod 4). In addition, divisors of the other forms need not be set equal to m^2-n^2 , $2mn$ and m^2+n^2 . Below is an example, where $x=30$.

$d_1=2$, $d_2=3$, $d_3=5$, $d_4=6$, $d_5=10$, $d_6=15$ and $d_7=x=30$.

d_1 is congruent to 2 modulo 4, so we need not consider it.

d_2 is congruent to -1 modulo 4, so we set $3=m^2-n^2$, which yields $m=2$ and $n=1$. These values produce the triple (3,4,5), which, when multiplied by $30/3$ gives (30,40,50).

d_3 is congruent to 1 modulo 4, so we first set $5=m^2+n^2$, which yields $m=2$ and $n=1$. These values produce the triple (3,4,5), which, when multiplied by $30/5$ gives (18,24,30). When we set $5=m^2-n^2$, we get $m=3$ and $n=2$. These values produce the triple (5,12,13) or (30,72,78).

"If you ask your mother for one fried egg for breakfast and she gives you two fried eggs and you eat both of them, who is better in arithmetic, you or your mother?"

--from "Arithmetic"
by Carl Sandburg

d_4 , d_5 and d_7 are congruent to 2 mod 4, so we need not consider them.

d_6 is congruent to -1 modulo 4, so we set $15 = m^2 - n^2$. This gives $m=4$ and $n=1$ as well as $m=8$ and $n=7$. These sets of values for m and n produce the triples (15,8,17) and (15,112,113), which, when multiplied by 30/15, become (30,16,34) and (30,224,226). Thus, the triples that contain 30 are:

(30,40,50), (18,24,30), (30,72,78), (30,16,34) and (30,224,226).

PROBLEMS submitted by Iuval Clejan

These problems actually appeared on past Junior meets. If you can do them, you should seriously consider trying out for the Math Team!

1. Express 999973 as a product of primes.
2. Three circles of radius 3 are all externally tangent to each other. A larger circle is internally tangent to all three circles, and a much smaller circle is externally tangent to the three circles. Find the difference in the areas of the largest and smallest circles.
3. Three adjacent faces of a box (rectangular parallelepiped) have areas 7, 14 and 18. Find the volume of the box.
4. The three lines whose equations are
$$\begin{aligned}3x - 5y &= 4, \\4x - ky &= 4, \\ \text{and } 5x - my &= 8\end{aligned}$$
all pass through the same point. Find $k+m$.
5. Two round pencils have diameters 6 and 8 millimeters. They are held together by a rubber band. Find the length of that part of the rubber band not in contact with the pencils.
6. In a circle, a 6-inch chord and a 14-inch chord are parallel and are 10 inches apart. The center of the circle is between the chords. Find the number of inches in the radius of the circle.

IN STOCK ALWAYS:
INTEGRATED CIRCUITS TTL, DIGITAL, LINEAR, ROMS,
RAMS, EPROM, MEMORY, TRANSISTORS, SOCKETS,
DIODES, BRIDGES, LED'S... KITS... FRIENDLY,
COURTEOUS SERVICE, LOW PRICES... BEGINNERS
WELCOME...

TRANS-AM ELECTRONICS
383 CANAL ST. N.Y. 10013
212-226-3893

Ptolemy's Theorem

by Enn-Ling Chen

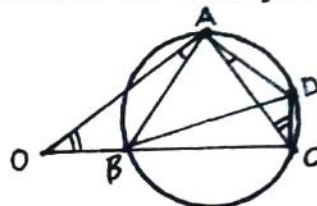
The cyclic quadrilateral is a topic which is not usually taught in secondary school geometry courses. A quadrilateral is said to be cyclic if all four of its vertices lie on the same circle. One of the most important theorems dealing with cyclic quadrilaterals is Ptolemy's Theorem, which we will now state and prove.

Ptolemy's Theorem: In a cyclic quadrilateral, the product of the diagonals is equal to the sum of the products of the pairs of opposite sides.

Proof of theorem: Draw $\overline{AO}$ so that $\angle BAO \cong \angle DAC$ and $\angle AOB \cong \angle ACD$.

Lemma: O lies on $\overline{BC}$ extended.

Proof of Lemma: It suffices to show that $\angle ABO \cong \angle ADC$. This is in fact the case because $\angle ABC$ and $\angle ADC$ are supplementary, as can be shown by noting that $\angle ABC = \frac{1}{2}\widehat{ADC}$ and $\angle ADC = \frac{1}{2}\widehat{ABC}$ and that $\frac{1}{2}(\widehat{ADC} + \widehat{ABC}) = 180^\circ$.



It is clear that triangles AOB and ACD are similar. Hence

$$AO:AC = AB:AD = OB:CD. \quad (1)$$

Furthermore, triangles OAC and BAD are similar because $\angle OAC \cong \angle BAD$ and $\angle AOC \cong \angle ABD$ (since $\angle ABD$ intercepts the same arc as $\angle ACD$, which is congruent to $\angle AOC$). Therefore,

$$OC:BD = AC:AD. \quad (2)$$

By the lemma,

$$OC = OB + BC. \quad (3)$$

From (1), $OB = (AB)(CD)/AD$, and from (2), $OC = (AC)(BD)/AD$.

Thus,
$$\frac{(AC)(BD)}{AD} = \frac{(AB)(CD)}{AD} + BC.$$
 Multiplying through by AD

gives the desired result.

Q.E.D.

This leads us to question the relationship between the product of the diagonals and the sum of the products of the pairs of opposite sides when the quadrilateral is not cyclic. It turns out that in this case, O does not lie on $\overline{BC}$ extended. Therefore, by the triangle inequality, $OC < OB + BC$. Making the same substitutions as we made earlier yields

$$(AC)(BD) < (AB)(CD) + (BC)(AD). \quad (4)$$

In other words, in a noncyclic quadrilateral the product of the diagonals is smaller than the sum of the products of the pairs of opposite sides. Thus we have proven the converse of Ptolemy's Theorem: If the product of the diagonals of a quadrilateral is equal to the sum of the products of the pairs of opposite sides, then the quadrilateral is cyclic.

Pascal's Triangle

by Mark Basch & Nir Hacoen

Pascal's Triangle is a fascinating triangular array of numbers with an astounding number of mathematical properties. Although it was known to the Chinese as early as the 14th century, the triangle is named after the 17th century mathematician and philosopher, Blaise Pascal, who discovered many of its features. The triangle is constructed according to a simple pattern: the two diagonal sides of the triangle consist of 1's, and each number in the interior is the sum of the two numbers above it (see below).

The primary property of the triangle is that the n th row consists of the numbers

$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \binom{n}{3}, \dots, \binom{n}{n}$, where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \text{ and } n! = n(n-1)\dots 3 \cdot 2 \cdot 1.$$

	Row
1	0
1 1	1
1 2 1	2
1 3 3 1	3
1 4 6 4 1	4
.	.
.	.

This can be verified as follows:

We wish to show that

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}. \text{ For example, } \binom{3}{2} + \binom{3}{3} = 3 + 1 = 4 = \binom{4}{3}.$$

$$\begin{aligned} \binom{n}{k} + \binom{n}{k+1} &= \frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{(k+1)n! + (n-k)n!}{(k+1)!(n-k)!} = \\ &= \frac{(n+1)n!}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n+1-k-1)!} = \binom{n+1}{k+1}. \end{aligned}$$

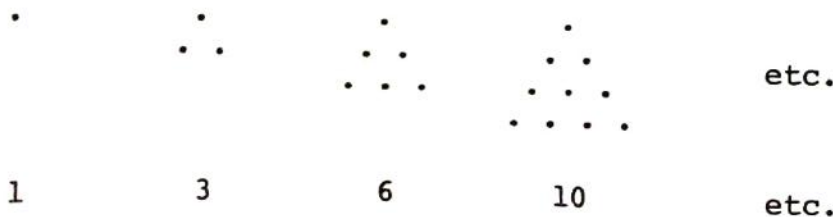
Since the binomial theorem states that

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{n}b^n,$$

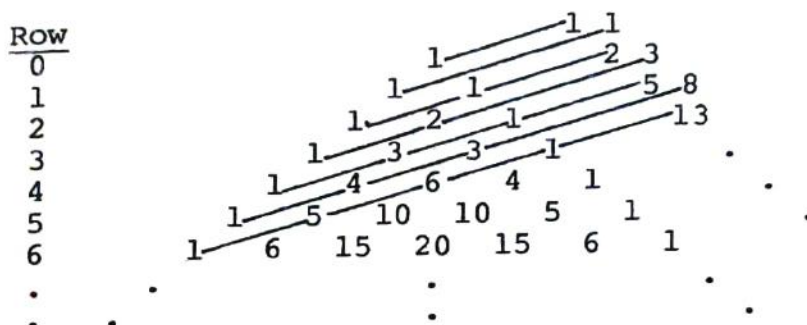
we can use Pascal's Triangle to expand polynomials. For example, to find $(x+2)^4$, we look at the fourth row, and find that

$$\begin{aligned} (x+2)^4 &= 1 \cdot x^4 + 4 \cdot x^{4-1} \cdot 2 + 6 \cdot x^{4-2} \cdot 2^2 + 4 \cdot x^{4-3} \cdot 2^3 + 1 \cdot x^{4-4} \cdot 2^4 = \\ &= x^4 + 8x^3 + 24x^2 + 32x + 16. \end{aligned}$$

Another interesting characteristic of the triangle is that the third diagonal (1, 3, 6, ...) is the sequence of triangular numbers. These are the numbers of objects that can be packed into a triangular array of integral side length.



One of the most fascinating properties of Pascal's Triangle that was not discovered until the 1800's is the existence of the Fibonacci sequence (1,1,2,3,5,8,13,..., where every term is the sum of the previous two) in the triangle. As shown below, the sums of the gently sloping diagonals are the Fibonacci numbers.



WESTINGHOUSE SALUTES THE WINNERS



The Winners of the 41st Annual Science Talent Search on the East Steps of the Capitol

OF THE 1982 SCIENCE TALENT SEARCH

In 1981-2, 950 seniors submitted complete entries to the Science Talent Search. This coming year, hopefully, there will be well over a thousand entrants. However, this goal cannot be achieved without the help of this year's juniors. Everyone in the 11th grade is urged to submit a project.

While a substantial amount of work is entailed, the rewards are well worth the effort. In addition to the scholarships and awards given to the winners (\$89,500 was distributed among the 40 winners in 1981), the benefits derived from working on a project are numerous. First, the project allows a student to become accustomed to research techniques and to the work habits needed to complete a piece of research. These skills are essential ones which need to be learned by anyone planning to do research. In addition, the completion of an original piece of research can be extremely gratifying, and can serve as the basis for further research. Also, the winners receive more publicity and national attention than most people receive during their entire lives.

These are only a few of the ways in which entering the Science Talent Search can be worthwhile. To apply, contact your department chairman and request a Data Blank. A copy of the rules is below:

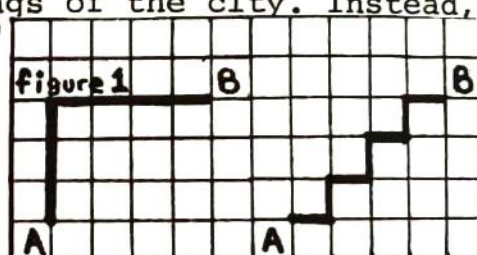
1. Each student must submit a report of about 1,000 words on an independent research project in the physical sciences, engineering, mathematics, biological sciences (excluding live vertebrate experimentation), or behavioral sciences.
2. Each entry must include a Personal Data Blank and secondary school record filled out by the student, his teachers and principal.
3. The secondary school record MUST include available national test scores, i.e., National Education Development Tests; Preliminary Scholastic Aptitude Test/National Merit Scholarship Qualifying Test; College Entrance Examination Board, etc.
4. No projects involving live vertebrate animal experimentation will be eligible. Projects involving human beings, and those on behavioral observations of animals in their natural habitat are excluded from this ruling and are eligible.
5. The project MUST be the work of a single individual. Group projects are not eligible.
6. A student who has attended a Summer Science Training Program and has used material from that experience as part of his project, MUST submit a statement from the Director of the SSTP giving descriptive and explanatory evidence of independence and creativity.

Taxicab Geometry

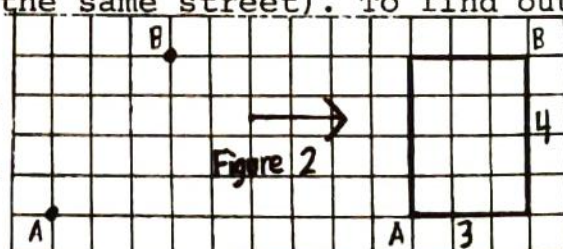
by Anna Franco

By changing one of the Euclidean geometry postulates, we can construct an entirely new type of geometry. The most famous non-Euclidean geometries are elliptic geometry and hyperbolic geometry. I will be discussing a lesser known and simpler type, an idea that was first proposed by the Russian mathematician, Herman Minkowski. It is called taxi-cab geometry because to understand it best, we pretend we are cab drivers and use graph paper to represent our streets and blocks. Taxi-cab geometry is derived by changing Euclid's first postulate which says that the shortest distance between two points is a straight line. This will no longer have to be true all the time.

A cab driver who wants to go from location A to location B could not possibly pass through the buildings of the city. Instead, the driver would use one or more "zig-zags" to get there (see figure 1). Therefore, instead of using the Pythagorean Theorem to find the distance between two points, we count the number of "blocks" driven by the cab. Thus, the distance between points A and B in figure 1 is not 5 but 7.



Along with this change come a number of other surprises. There is no longer a unique "shortest-distance" path between two points (provided that the points are not on the same street). To find out how many paths there are, we first draw a rectangle with these two points as opposite vertices (see figure 2). Then we use the formula used to find out how many ways one can select r objects from n objects. The formula is

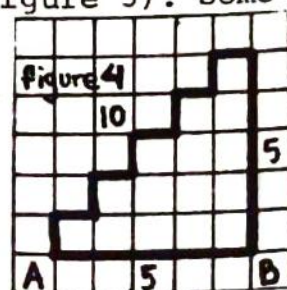
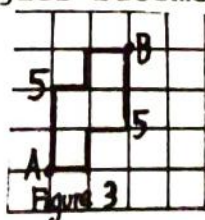


$$W = \frac{n!}{r!(n-r)!}, \text{ where } n \text{ denotes the taxi-cab distance}$$

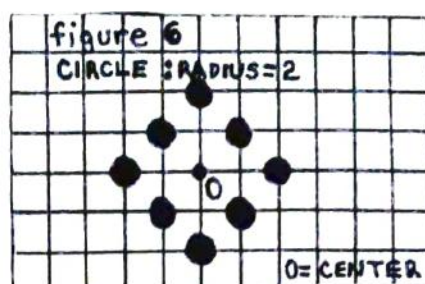
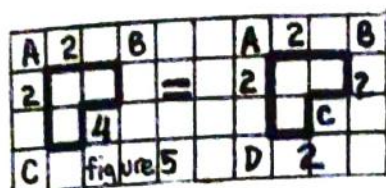
between the two points and r stands for one of the sides of the rectangle. Therefore, the number of paths from point A to point B in figure 2 is shown below.

$$n=7 \text{ and } r=3, \text{ so } W = \frac{7!}{3!(4!)} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 35.$$

Polygons become almost indistinguishable from each other, and angles become meaningless. However, we can draw biangles in taxi-cab geometry, while they are non-existent in Euclidean geometry. A biangle is a figure with two equal sides that join at two points (see figure 3). Some non-Euclidean triangles no longer follow the rule that says that the sum of any two sides must be greater than the third side. In figure 4, isosceles triangle ABC has sides of 5, 5 and 10. Obviously the sum of 5 and 5 is not greater than 10. A triangle can even be the same as a square, as



shown in figure 5.



The taxi-cab circle changes quite interestingly. As in Euclidean geometry, it is defined as the locus of points which are equidistant from a given point. Using this definition, the circle is easy to draw. (see figure 6).

Since the circle resembles a square, we can show that the equivalent of π in taxi-cab geometry is four. π is the ratio of the circumference of a Euclidean circle to its diameter. Since in taxi-cab geometry, the diameter equals the side of the circle, and the circumference of a Euclidean square is four times its side, we can say that " π " is equal to four.

The theorem which states that two circles can only intersect in two or fewer points no longer holds. In taxi-cab geometry, two circles can intersect in any number of points (see figure 7).

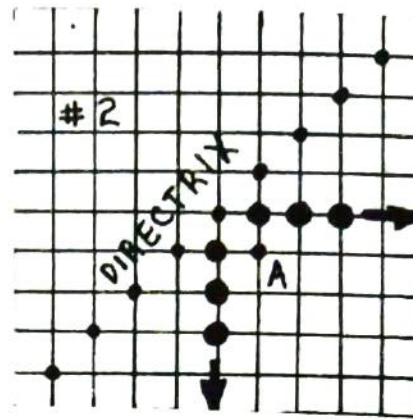
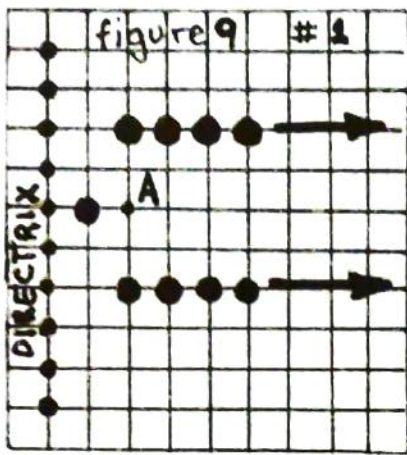
Let us consider what happens when we try to graph an ellipse. An ellipse is defined as the locus of points for which the sum of the distances to two given points remains constant. The given points are called foci.

There are four different types of ellipses (see figure 8).



The constant sum in the above figure is equal to 4, and the foci are labelled A and B. The first two ellipses have their foci on the same "street." Ellipse number 2 corresponds to the degenerate form of the Euclidean ellipse (which occurs when the distance between the foci is equal to the constant sum). Ellipse number 4 is the degenerate form of the ellipse when the foci are on different streets.

Given a focus, A, and a straight line called the directrix, a parabola is the locus of points whose distance to the focus is equal to the shortest distance to the directrix. If we define the directrix as a set of points from the Euclidean line, we can draw the taxi-cabian parabola (see figure 9 on the next page).



Once the fundamental differences between Euclidean and taxi-cab geometry are known, one can go on and try drawing hyperbolas and more complicated functions. It would also be interesting to see how taxi-cab geometry behaves in the third dimension. There has been only one book and a few articles written on taxi-cab geometry, and many questions about this topic still remain unanswered.

For more about taxi-cab geometry, see:

Martin Gardner, *Mathematical Games*, in *Scientific American*, November, 1980, pp. 18-29.



STUYVESANT HIGH SCHOOL'S

MATH SURVEY

PRESENTS

A CONTEST!

A Splendid Prize will be awarded to the entrant with the best complete and correct solutions to at least two of the four below problems. Members of the Math Survey staff are ineligible.

GOOD LUCK!!

PROBLEMS

1. The inscribed circle of acute triangle ABC is tangent to sides AB, BC, and CA at C' , B' , and A' , respectively. Draw triangle $A'B'C'$, and let its inscribed circle be tangent to its sides at A'' , B'' and C'' . Continue forming triangles in this fashion: $A''B''C''$, $A'''B'''C'''$, etc. Find and prove a construction for the limit point of the sequence of triangles ABC, $A'B'C'$, $A''B''C''$, etc.
2. "Find me a number," said the King to his jester, "half of which is a square." "How simple," said the jester. "But," continued the King, "one third of it must be a cube." The jester looked serious for once. "And finally," remarked the King, "one fifth of it must be a fifth power." The jester looked crestfallen. "Solution by tomorrow morning," finished the King, "or you must marry my widowed mother-in-law." The jester was crushed. But the next morning he had the answer and was saved from a fate worse than death. Can you find the number?
3. The following are examples of chains of lengths 4 and 5 respectively:

25, 225, 1225, 81225;
25, 625, 5625, 75625, 275625.

In each chain, each link is a perfect square, and each link (after the first) is formed by prefixing a single digit to its predecessor. For which values of $n \geq 6$ do there exist chains of length n ?
4. Can the product of k consecutive integers ever be a perfect square? ($k > 1$)

SEND YOUR SOLUTIONS TO:

DEADLINE: July 17, 1982

MATH SURVEY CONTEST
c/o Joel Hirschhorn, Editor-in-chief
29 Washington Square West
New York, N.Y. 10011